Modeling progress: causal models and the imperfective paradox

Elitzur Bar-Asher Siegal
Hebrew University of Jerusalem
ebas@mail.huji.ac.il

Prerna Nadathur
The Ohio State University
nadathur.3@osu.edu

July 31, 2025

Abstract

Durative telic (*accomplishment*) predicates are usually taken to denote a set of complete eventualities, crucially including the realization of a lexically-specified *culmination condition*. This assumption is in tension with the observation that accomplishment predicates are readily used to describe **non-culminated** events in a range of grammatical contexts: most famously, in the scope of the *progressive* aspect. Prominent accounts of this **imperfective paradox** (Dowty, 1979) tie the effect to the modal accessibility of culmination, intensionalizing the progressive operator so that it instantiates culminated events across a set of alternatives to the evaluation world. We argue that this approach faces empirical challenges, and in particular fails to capture the acceptability of telic progressives of *unlikely events* and locally *out of reach* tasks. We use these data, combined with insights from extensional treatments of the paradox (Parsons, 1989) to motivate a new approach, on which telic progressives are sensitive to part-whole structure inherited from an **event type** associated with the underlying predicate. On our analysis, an event type corresponds to a **causal model** (Pearl, 2000) in which the culmination condition C_P of P occurs as a dependent variable. The model provides a set of *causal pathways* for realizing C_P, each of which comprises a set of jointly sufficient causal conditions for C_P; it also establishes (sets of) conditions which preclude C_P. This approach allows the truth and felicity of telic progressives to depend not on (modal) expectations of culmination, but instead on a (partial)s match between reference time facts and causal pathways in the event type: concretely, a reference situation s verifies that an event of type P is *in progress* just in case it satisfies a non-trivial subset of the conditions in some causal pathway for C_P (indicating that a process for culmination has been initiated), and fails to verify a sufficient set of conditions for non-culmination (leaving open the possibility of further development towards C_P). While our approach complexifies the denotation of eventuality predicates, it offers a simplified *partitive* perspective on grammatical aspects. Looking ahead, the account suggests a deep link between lexical aspectual properties and the formal structure of causal models which promises to shed light on (non-)culmination phenomena beyond the progressive context.

1 Introduction

The prevalent approach to verb meaning in contemporary lexical semantics is one of *predicate decomposition* (Hale and Keyser, 2002; Grimshaw, 2005; Levin and Rappaport Hovav, 2005, a.o.). This type of theory divides verb meaning into two components, one structural and the other idiosyncratic. The first component characterizes *event type*, dividing verbs into distinct semantic clusters or *classes* on the basis of the inherent internal structure of the events they describe, and feeding the grammatical properties of the verb. The idiosyncratic component—the *root*—differentiates verbs within a given structural class, imparting specific but grammatically inert semantic content.

Within this approach, roots are systematically linked to event structures: each root has an ontological classification drawn from a fixed spectrum of types (e.g., *state, result state, place/container, location, manner, ...*) which determines where and how it can be integrated into a given event schema. The resulting representations are typically named by their roots, while the event scheme—expressed in terms of combinations of primitive predicates such as ACT, CAUSE, BECOME (Dowty, 1979)—classifies the verb as either simple (comprising a single subevent) or complex (comprising two subevents; Levin and Rappaport Hovav 1999). Example (1) provides simple schema, corresponding respectively to *activities, states,* and *achievements* (Vendler 1957). Complex events may be analyzed as *causative*, linking two simple structures by means of a CAUSE relation, as in (2). The transitive verb *open* could then be given the decompositional representation in (3).[1]

(1) Simple event schema:
  a. $[\,x\ \mathsf{ACT}_{\langle \mathit{MANNER}\rangle}\,(y)]$
  b. $[\,x\ \langle \mathit{STATE}\rangle\,]$
  c. $[\ \mathsf{BECOME}\,[\,x\ \langle \mathit{STATE}\rangle\,]\,]$

(2) Complex event scheme
  $[\,[\,x\ \mathsf{ACT}_{(\langle \mathit{MANNER}\rangle)}\,]\ \mathsf{CAUSE}\,[\,\mathsf{BECOME}\,[\,y\ \langle \mathit{RES\text{-}STATE}\rangle\,]\,]\,]$

(3) *open*: $[\,[\,x\ \mathsf{ACT}\,]\,\mathsf{CAUSE}\,[\,\mathsf{BECOME}\,[\,y\ \langle \mathit{OPEN}\rangle\,]\,]\,]$

The complex causal schema in (3) has three important features:

(A) Only the second argument of the CAUSE relation (i.e., the result event) contains an idiosyncratic component.[2] The participant argument of the first event interacts solely with the primitive predicate ACT. This aligns well with the apparent non-specificity of an implied causing action for change-of-state verbs such as *open*: for instance, capturing the intuition that (3) expresses its subject's responsibility for the specified result state without constraining the method in which x acted on y to bring this state about.

[1] Primitive (non-idiosyncratic) predicates are indicated in (1)-(3) without italicization; roots and ontological types are shown in angle brackets and italicized. Roots can occur as modifiers of primitive predicates (indicated in subscript, following Levin and Rappaport Hovav 2005) or as arguments. All optional elements are indicated parenthetically.

[2] The decomposition in (3) is characteristic of causative *change of state* verbs (*open, break, empty, ...*), and our discussion focuses on this verbal class. We do not mean to suggest that accomplishments generally lack an idiosyncratic manner component; on the contrary, predicates such as *run a marathon* or *bake a cake* clearly specify a modifying manner as part of their 'process' (i.e., the ACT component of 2).

The role of manner specifications in the decomposition of complex events is closely tied to the *Manner/Result Complementarity Hypothesis* (MRCH; Levin and Rappaport Hovav 1991), which posits that verbs lexicalize at most one of the two components: *manner* or *result*. If the MRCH holds, verbs that encode complex events must lexicalize a result component, thereby precluding the presence of a manner component (see Levin and Rappaport Hovav 2013, 2014; Beavers and Koontz-Garboden 2012 for a range of perspectives on this hypothesis).

We do not commit to a strong position regarding the MRCH. For our purposes, what matters is the frequent absence of a specified idiosyncratic component in the first subevent. As shown by Rappaport Hovav (2017), even causative change-of-state verbs like *drown*, which sometimes appear to involve manner entailments, can be used in contexts where no specific manner is implied—e.g., when the subject "didn't move a muscle." Such cases suggest that these verbs can describe result states brought about without lexical specification of the causing action, supporting our claim that their truth conditions do not hinge on a particular manner or idiosyncratic action leading to the result.

(B) Per the inclusion of CAUSE, some analysis of causal relationships must underpin the connection between the argument of ACT and the predicate's result state. Neither (2) nor (3) commits to the nature of the causal relation, taking CAUSE as a (lexical) semantic primitive.

(C) The inclusion of both CAUSE and an associated result state in (2)/(3) gives rise to what we call the **culmination assumption:** that the denotation of an uninflected telic predicate exclusively contains eventualities which describe a complete developmental trajectory from some point of initiation up to and including the realization of the result state—i.e., a point of culmination, beyond which the event cannot continue.

Complex predicates of the sort schematized in (2) are often categorized as **durative telic predicates** (Vendlerian *accomplishments*), in view of their association with a *process* (leading from initiation to termination) as well as a point of *culmination.* These culmination points represent categorical transitions between the developing eventuality and the (root-specified) result state. The outcomes or products of such culminations are thus precisely delineated by the decompositional representation of the underlying verb. These outcomes can range from the creation or destruction of an object (see 4a, 4b, respectively) to arrival at the terminus of a path (4c), to an instantaneous transition between states (of some pre-existing object, as in 4d).

(4) a. Emanuel baked a cake.
result: A (baked) cake. *object creation*

b. Maya ate a cookie.
result: A consumed cookie. *object destruction*

c. Des ran a marathon.
result: A complete (running) traversal of a 26.2 mile path *path termination*

d. Liina opened the door.
result: An open door. *state transition*

Prima facie, the schema in (2)—and the culmination assumption in particular—are in tension with the cross-linguistically robust observation that accomplishment predicates can be felicitously used to describe non-culminating eventualities. Contexts permitting non-culminating instantiation include, for instance, the complements of certain *aspectual verbs*, as shown for English in (5):

(5) a. Emanuel began to bake a cake.

b. Emanuel continued to bake a cake.

c. Emanuel stopped baking a cake.

Two things are immediately clear. First, none of (5a)-(5c) depend for their truth on the eventual existence of a complete (baked) cake. (5a), for instance, felicitously describes a situation in which Emanuel mixed together butter and sugar with the intention of making a cake, but went on to discover that the butter was rancid, putting a stop to the endeavour; (5b) might describe a situation in which he proceeded from some set of activities satisfying (5a) to a more advanced stage of the baking procedure (e.g, by adding flour and eggs to the butter-sugar mixture), but is compatible with his then being called permanently away for some emergency. Similarly, (5c) might narratively serve to introduce the emergency, as long as Emanuel's prior activities satisfy at least (5a).

Second, despite the apparent irrelevance of culmination, truth conditional assessment of (5a)-(5c) is linked to some relationship between Emanuel's reference time actions and the culmination specified by the embedded predicate. This relationship is what binds situations verifying (5a)-(5c) together as (partial or incomplete) instances of cake-baking, and what distinguishes these situations from activities like wiping counters, answering the phone, or chopping vegetables. Intuitively, situations validating (5a)-(5c) must plausibly be *parts* of a recognizable cake-baking process: that is, they must resemble the activities one might describe when asked to explain how to bake a cake.

The trouble is this: nothing in a decompositional representation such as (2) appears to supply the sort of rich procedural information required to cash out the descriptive generalization above. While the acceptability of (5a)-(5c) is of course partially governed by the lexical semantics of the embedding verbs (*begin* selects for initiating activities, *continue* for intermediate stages, and so on), these embedding predicates must operate on the bipartite causal structure in (2). The 'procedural' part of a complex eventuality is specified in terms of structural (non-idiosyncratic) components: ACT (and its arguments), and CAUSE. As a result, any characterization of qualifying ACT eventualities must be constrained only by their causal relationship to the result state: moreover, where the result state itself goes unrealized, the CAUSE relation cannot itself be fully realized, leaving the set of situations which plausibly verify claims like (5a)-(5c) apparently unconstrained.[3]

This is evidently untenable. It is clear that certain activities (mixing butter and sugar together) are potential truth-makers for (5a)-(5c), while others (taking a phone call) are not. It is equally clear, moreover, that the information that determines whether or not a particular set of activities 'counts' as any part of a cake-baking is introduced by the predicate itself (and not by the embedding context). The preceding discussion thus illustrates an important point: truth conditional assessment of (non-culminating) uses of telic predicates relies on the availability of a rich body of *world knowledge* pertaining to the typical processes and procedures that bring about the appropriate culmination (and/or result state). While this information is evidently tightly linked to the specified culmination, it is—equally evidently—not itself made explicit by lexical means. The resulting puzzle of telic predicates is twofold. First, how is the world knowledge relevant for linguistic assessment mediated through the specified culmination? Second, by what formal means does this information enter into truth-conditional assessment? How these questions are answered depends on how the notion of a process leading to a certain result is to be captured and, consequently, on how *progress towards culmination* is to be recognized and formally assessed.

This paper aims to address the above challenges by introducing a new approach to the semantic structure of telic predicates, challenging the conventional decompositional analysis presented in (2). We critically (re)evaluate implications (A)-(C), making the following claims.

(I) Against the (widely-accepted) culmination assumption (see C above), we argue that both culminating and non-culminating eventualities can validly instantiate telic predicates, as long as they exhibit an appropriate formal alignment with recognizable procedures for culmination.

(II) The representation of a telic predicate P does not hinge on a culmination *requirement*. It centers instead on a **culmination condition:** a specification of what can or must be true for an event of type P to culminate (cf. Kratzer 2004). Thus, unlike the decompositional scheme in 2 (see also A above), the predicate must—in addition to specifying a relevant result state—provide idiosyncratic information concerning what leads to this result. In other

[3]The problem is specific to complex eventualities, due to the structure of structural vs. idiosyncratic components in the proposed decompositions. As schematized in (1), predicates denoting simple events allow for interaction between the idiosyncractic root and relevant semantic primitives, so that contexts which reference the initiation or development of a simplex predicate have direct access to idiosyncratic information.

words, a telic predicate feeds truth conditional evaluation a structured body of information about the *causes* of culmination. This approach requires an account of causal relationships that goes beyond a binary primitive such as CAUSE (see B above).

Concretely, we propose that the event type associated with a telic predicate P corresponds to a formal **causal model** (cf. Pearl 2000) in which P's culmination condition, C_P, occurs as a dependent variable. In the abstract, such a model comprises a set of **causal pathways** (informally: procedures; formally: sets of collectively sufficient causing conditions) for the realization of C_P, as well as pathways for its failure. The precise content of these pathways is provided by observation and/or experience, so that an event type model imposes set-theoretic structure on the world knowledge that constitutes our sense of how to bring C_P about: that is, on the sort of world knowledge that is crucial for truth-conditional assessment of (non-)culminating instantiations of P. Within the structure of an event type model, both partial and culminated instantiations of a telic predicate can be defined in terms of subset comparison to a complete causal pathway for C_P.

This paper thus advances causal models as cognitive objects that bridge the gaps between lexically-specified information, world knowledge, and the set-theoretic objects on which a compositional semantic system is understood to operate. We will motivate the causal approach to telic predicates by means of a well-known case study in non-culmination: the so-called **imperfective paradox** of telic progressives (Dowty 1979). By revising the decompositional representation in (2), and letting go of the associated culmination assumption, our approach obviates the paradox, allowing us to assign an intuitive *partitive* semantics to the progressive operator.

The paper is structured as follows. Section 2 provides background on telic non-culmination in the progressive context, focusing on a set of data which challenge existing analyses of the imperfective paradox. In section 3, we introduce structural equation causal models, and show how they can be fruitfully used to explicate truth-conditional linguistic judgments: we offer a formal definition for telic predicates within this framework. Section 4 returns to telic progressives, offering a compositional treatment which handles the outstanding data from Section 2. We conclude in Section 5 by summarizing our account of the imperfective paradox, and discussing its broader consequences.

2 Culmination, expectations, and partial realization

2.1 Background: the imperfective paradox

The lexical puzzle of telic predicates has primarily been viewed through the lens of the **imperfective paradox**, which centers on the interpretation of accomplishment predicates under progressive modification (Dowty 1979). Unlike atelic predicates, which exhibit a symmetric pattern of entailment between their progressive and nonprogressive forms (as shown in 6), telic progressives do not entail their nonprogressive counterparts (Kenny, 1963).

(6) Atelic (activity) predicate:

a. *Simple past (perfective):* Lila pushed a cart. $\leftrightarrow$

b. *Progressive:* Lila was pushing a cart.

(7) Telic (accomplishment) predicate:

a. *Simple past (perfective):* Lila drew a circle. $\rightarrow$

b. *Progressive:* Lila was drawing a circle.

Intuitively, the lack of entailment from (7b) to (7a) is due to the distinct relationship that progressive and nonprogressive aspects have to the culmination condition of an uninflected accomplishment predicate. (7a) presents a completed picture, conveying that culmination occurred within reference time, and thus entailing the existence of a complete culmination product (i.e., a full circle, drawn by Lila). Progressive (7b) has no such requirement: like the aspectual verb constructions in (5a)-(5b), it requires only that some part of a circle-drawing coincides with reference time, so that Lila's actions *make progress* towards culmination but need not get all the way there.

While there is a clear contrast with (6), it is not immediately obvious that the asymmetric entailment pattern in (7) constitutes a 'paradox'. As formulated by Dowty (1977, 1979), the puzzle emerges from a clash between the empirical acceptability of telic progressives in contexts which preclude culmination, and standard assumptions about (a) the denotation of telic predicates, and (b) the contribution of grammatical aspects. As noted in the introduction, the prevailing **culmination assumption** is that the complex eventualities denoted by telic predicates necessarily incorporate the point of culmination. This perspective gains support from the truth conditions of perfective accomplishments in a range of languages (exemplified for English by the simple past in 7a), which license reference time **culmination entailments**.

Culmination entailments may be readily explained by combining the culmination assumption with an extensional treatment of the perfective aspect (see 8), on which the 'completed-event' perspective is captured by including the runtime ($\tau(e)$) of an instantiated eventuality (e) in the reference time (t) supplied by tense.[4] Since, per the culmination assumption, e qualifies as a P-event just in case it includes a realization of P's culmination condition (C_P), perfective modification forces a reference time realization of C_P. The culmination entailment follows immediately:

(8) $[\![\textsc{pfv}]\!] := \lambda w \lambda t \lambda P_{vt}.\exists e[\tau(e) \subseteq t \wedge P(e)(w)]$ (Klein, 1994; Kratzer, 1998)

(9) $[\![(7a)]\!]^{w^*,t^*} = \left[\!\left[\textsc{pst}(\textsc{pfv}(\sqrt{}\text{Lila draw a circle}))\right]\!\right]^{w^*,t^*}$
$= \exists e[\tau(e) \subseteq t^*\{\prec \mathsf{now}\} \wedge \mathsf{draw\text{-}circle(L)}(e)(w^*)]$

In contrast to the completed ('bird's-eye') perspective of the perfective, progressives portray a given situation as *ongoing* (in progress) at reference time. Extrapolating from above, then, a natural semantics for the progressive might reverse the key inclusion relation in (8) to yield (10).

(10) $[\![\mathsf{PROG}]\!] := \lambda w \lambda t \lambda P_{vt}.\exists e[\tau(e) \supset t \wedge P(e)(w)]$

(11) $[\![(7b)]\!]^{w^*,t^*} = [\![\textsc{pst}(\mathsf{PROG}(\text{Lila draw a circle}))]\!]^{w^*,t^*}$
$= \exists e[\tau(e) \supset t^*\{\prec \mathsf{now}\} \wedge \mathsf{draw\text{-}circle(L)}(e)(w^*)]$
entails: $\exists t'[t' \supset \tau(e) \wedge \mathsf{draw\text{-}circle(L)}(e)(w^*)]$

Combining the culmination assumption with (10) thus straightforwardly predicts that the progressive of a telic predicate P can only be true if reference time events continue to develop to culmination. But this directly contradicts the empirical data: as demonstrated by (12), telic progressives can be both acceptable and true in contexts which rule out any possibility of culmination.

(12) Mahler was composing his tenth symphony (when he died). (True)

[4]This formulation assumes a three-times system (Reichenbach, 1947; Klein, 1994), on which aspects relate event time to reference time, and pronominal tenses (Partee 1973; Kratzer 1998) establish a relationship between reference time and speech time. Events (type v) and times (type i) are related via Krifka's (1989) *temporal trace* function, $\tau : D_v \rightarrow D_i$, with $\tau(e)$ returning the runtime of an input eventuality e.

Since Dowty's 'paradox' follows from combining the culmination assumption with an extensional treatment of (the progressive) aspect, the two obvious routes towards its resolution involve modifying one or the other of these assumptions. Both approaches—revising the extensional progressive in (10) and relaxing the culmination assumption—have been explored in the literature, and we summarize the key ideas of each approach below.

(I) **Type I: Intensional progressive approaches**
The culmination assumption can be preserved on an approach which intensionalizes the progressive operator (see Dowty, 1977, 1979; Landman, 1992; Asher, 1992; Bonomi, 1997; Zucchi, 1999, a.o.): the idea is that the progressive instantiates qualifying (i.e., culminated) P-eventualities in a set of alternatives to the evaluation world. On this approach, $\mathsf{PROG}(P)$ is true of a situation s at time t in world w just in case s continues to develop up to the realization of C_P in some set of worlds which look sufficiently similar to w at time t.

The central analytical challenge is to appropriately constrain the alternatives introduced by PROG. This set must plausibly exclude the evaluation world, in order to allow for data like (12). Simultaneously, however, the alternatives must track actual events fairly closely, in order to capture the intuition that the reference time facts correspond to some initial portion (or *stage*; Landman 1992) of a P-event—i.e., to establish that a (potentially incomplete) P-event is actually *in progress* in w at reference time.

(II) **Type P: Extensional or partitive approaches.**
Our other option is to set aside the culmination assumption, allowing (non-)culminated eventualities to validly instantiate a telic predicate (Parsons 1989, 1990; see also Lascarides 1991; Szabó 2004, 2008). In principle, this move obviates the imperfective paradox, allowing us to account for data like (12) while maintaining an extensional treatment of the progressive.[5]

Type P approaches must address the following question: what constitutes a valid *partial realization* of a telic predicate? In the context of (12), for instance, a satisfactory answer must explain what attributes of an unfinished symphony-writing activity align it with a completed instance of the same predicate, while distinguishing it from seemingly comparable occurrences of a related but atelic activity (e.g.,*composing music*).[6]

Our view is that the two analytical challenges outlined above are intertwined. Whether one adopts an extensional or intensional analysis of the progressive operator, truth conditional assessment of 'paradoxical' data hinges on the concept of *partial realization*. This notion must be a partitive one, insofar as reference-time facts can only constitute a partial (non-culminated, process-only) P-type event if there is some abstract or concrete whole of which these facts form a well-defined part. At the same time, partial realization (at least in the progressive context) must have an intensional quality: an *in-progress* P-event must naturally reflect a progression of facts that has advanced some way towards a (not yet realized but in some sense realizable) culmination condition.[7]

[5] Although the perfective is not our focus here, it is worth noting that Type P approaches must offer a different explanation than temporal containment (the contrast presented by 8 and 10) to capture the differences between perfective and progressive aspects. See Section 4 for an additional comment.

[6] From the Type P perspective, the imperfective paradox is more appropriately termed the *partitive puzzle* (Bach 1986); its solution extends beyond a treatment of telic progressives and promises to explain the interpretation of telic predicates in other non-culminating contexts (such as those exemplified in 5).

[7] To the extent that an in-progress or non-culminated instantiation of a telic predicate P describes exclusively a

Looking ahead, we use causal models to combine the partitive and intensional perspectives in a general theory of telic predicates. This will enable us to provide a clear definition of the abstract, culmination-linked whole of which an in-progress P-eventuality must be a component. Unlike standard Type I analyses—which introduce intensionality exclusively via the progressive operator—the causal-modeling approach builds the crucial intensional component of the imperfective paradox directly into the denotation of a telic predicate. As a result, we avoid making the 'partiality' of reference-time activities contingent on the locally-projected likelihood of culmination. This will allow us to explain some key data that has persisted as a challenge to the received Type I approach.

Before introducing our formal proposal, we review some key developments in Type I approaches, focusing our discussion on the challenges posed by three types of data: progressives associated with **impossible events**, **unlikely events**, and scenarios in which culmination is locally but not categorically **out of reach**. We use these data to lay the groundwork for an approach that integrates a guiding principle of the Type P approach: namely, that reference time facts matter more than modal projections in determining whether or not a given situation constitutes a (valid) partial realization of a telic predicate. We conclude this section with a new descriptive generalization for the truth conditions of telic progressives, which will form the basis of our formal account.

2.2 Inertia and expectation

From the Type I perspective, a *partial realization* of a telic predicate P is a situation which 'naturally' leads to the realization of P's culmination condition C_P. $\mathsf{PROG}(P)$ is true of a situation s just in case s continues to develop until the realization of C_P in some set of normal or typical alternatives to the evaluation world. This approach is intuitively appealing to the extent that it captures the idea that an in-progress (or process-only) P-eventuality is one which 'aims' at culmination. However, it has proven difficult to characterize the relevant set of alternatives in a way that properly distinguishes acceptable from unacceptable uses of telic progressives.

Dowty's (1979) influential analysis aims to cash out an appropriate modal relationship in terms of *inertia*. Reference-time activities are taken to verify $\mathsf{PROG}(P)$ just in case they eventually realized C_P if allowed to continue uninterrupted. The set of *inertia worlds* $\mathsf{INR}(w,t)$ projected from w at time t are defined as those worlds which share a history (see Thomason 1984) with w through t, and which are further constrained to develop after t in the way that is "most compatible with the past course of events" (Dowty, 1979, p.148): the result is that a (future) realization of C_P is treated as the normal outcome of world history as considered from the reference time perspective.[8]

(13) **The inertia world analysis** Dowty (1979, p.149)
For a telic predicate P, $\mathsf{PROG}(P)$ is true at $\langle w,t\rangle$ iff, for some interval t' such that t is a nonfinal subinterval of t', and for all worlds $w' \in \mathsf{INR}(w,t)$, P is true at $\langle w',t'\rangle$.

Dowty's reliance on a global perspective (i.e., one which takes an entire world into account) has undesirable consequences (Vlach, 1981): linking inertia worlds to the complete history of the evaluation world incorrectly predicts the falsity of telic progressives in any context where an event which

process component of P, the tension between Type I and P approaches (and even between different theories of each type) hinges on how the notion of a process for culmination should be understood. Although we do not uniformly frame our discussion of progressives in terms of the process concept, our formal proposal takes a clear position on how *process* should be understood; see the appendix for further discussion.

[8] Compatibility is based on Lewis's (1973b; 1979) *similarity*, which is related to *stereotypicality* (Kratzer, 1981). The type of information needed to assess stereotypical development—that is, that guides expectations about the *normal developments* of a situation—is very plausibly causal in nature (see, e.g., Kaufmann 2013; Nadathur 2023a).

actually blocks culmination is foreseeable at reference time. This makes the wrong prediction for (14): since $\mathsf{INR}(w,t)$ picks out worlds in which all reference time a activities continue as established, it includes worlds which contain both the inertial continuation of Henrietta's trajectory, as well as that of the offending truck. As a result, the fatal collision must occur across $\mathsf{INR}(w,t)$. This should immediately falsify (14a), but the claim is judged to be both acceptable and true in context.

(14) **The collision scenario.** Henrietta walked into a crosswalk, intending to cross to the other side of the street. As she took her first step into the street (but unknown to her), a truck was racing towards the same crosswalk with a speed and trajectory that placed it on a collision course with Henrietta (given her speed and trajectory).

a. *Target:* Henrietta was crossing the street (when the truck hit her).

The problem can be avoided by refining Dowty's proposal so that the information used to project inertial futures is restricted to some subset of the global picture. One way of doing this is to allow the inertial development of only those processes which are understood to belong in some sense to the target predicate (cf. Landman 1992; Bonomi 1997; see Section 4.3).[9] Alternatively, the same basic intuition can be captured by restricting attention to a particular *perspective* on the reference time situation, roughly constituted by a subset of the (global) facts at reference time (Asher, 1992).

Asher's account is technically complex, but its central idea can be expressed fairly simply. Intuitions about the role of normal or inertial developments are expressed in terms of *default* inferences, concerning what typically follows from a particular body of information. On this view, the progressive of a telic predicate P can hold at time t just in case there is some *admissable perspective* π on the full set of reference time facts (i.e., some suitably-constrained subset of these facts) such that all of the maximally normal worlds in which π holds are worlds in which the nonprogressive form of P holds at or after reference time (s normally leads to C_P).

In addition to explaining why a telic progressive can hold in 'predictable failure' contexts such as (14)—the perspective adopted here is ostensibly that of Henrietta herself, who is not aware of the oncoming truck—Asher's restricted normality-based approach also explains the observation that progressives of telic predicates with (locally) contradictory culmination conditions can be true of the same general state of affairs. Both (14a) and (15) are truthful descriptions of the context in (14), even though worlds in which the culmination associated with (15) occurs are, by definition, worlds in which Henrietta does not reach the other side of the street (Abusch, 1985).

[9] Landman (1992) calls this the *normality proposal*; the idea is to relativize the set of inertia worlds to an event e as well as a world-time pair, so that a world w differs from its inertial alternatives with respect to e at most in that e follows its 'natural course' in any inertia world. This differs from Vlach's (1981)'s solution insofar as Vlach links the truth of a progressive to the *causal consequences* of reference-time events, rather than their normal developments. Landman argues that Vlach is insufficiently specific about the parameters under which events in progress are assumed to continue, pointing out for instance that if the progressive in (14a) is used at the moment when Henrietta steps into the crosswalk, the evaluation world is one in which the crossing process continues past reference time, but nevertheless is not a world in which this process leads to the truth of the corresponding non-progressive. Thus, for Landman, Vlach's proposal must be refined to require that the ongoing process continues *past the point at which it is interrupted in* w^*, and this forms part of the basis for his own proposal (see Section 4.3).

The causal component of Vlach's proposal is amenable to our own view. However, it remains subject to an objection due to Bonomi (1997) (leveled at Asher 1992): the mere existence of a causal relationship between reference time activities and the culmination condition of predicate P is not enough to license $\mathsf{PROG}(P)$. For instance, Henrietta's waking up in the morning may eventually lead to her crossing the street in (14a); nevertheless, neither (14a) nor its present tense correlate can be valid at a time before that at which Henrietta first sets foot in the street. Thus, while causal relationships between reference time activities and C_P are truth-conditionally relevant for $\mathsf{PROG}(P)$, a causal analysis remains incomplete without an accompanying mereological conception of the target event type.

(15) Henrietta was walking to her death.

Asher's explanation is straightforward: "it suffices for the truth of the progressive that there be just one perspective π on [a state of affairs] s such that the normal course of events based on having a state with the characteristics given in π leads to a completion of the appropriate kind" (1992, p.480). Thus (14a) can be true because s contains a set of facts pertaining only to Henrietta and her activities which licenses a default inference to her eventual successful crossing; however, since s also contains a (larger) alternative set of facts including the truck, there is an alternative perspective on the situation which licenses a default inference to the fatal collision.

Generalizing, Type I approaches tie the truth conditions of telic progressives to the validity of some (local) expectation of culmination, while allowing this expectation to be based on what is in some cases an extremely narrow construal of the evaluation world facts (see also Lascarides 1991 on the "eventual outcome strategy"). This seems to capture the observed unacceptability of progressives of **impossible tasks**, such as those in (16):

(16) a. #/?Mary is swimming across the Atlantic Ocean.
b. #/?The children are digging a hole to China.

It is certainly possible for Mary to get into the ocean at Galway and begin to swim generally westward, but there is no realistic perspective which admits both Mary's basic human capabilities and the properties of the Atlantic Ocean while simultaneously predicting successful crossing as the typical outcome of her reference time actions. Similarly, for (16b), while it might be true that some children are digging a hole in the sand at place on the Earth's surface which is antipodal to China, no perspective which admits basic facts about the earth's composition, physics, and the limits of human strength and heat tolerance will license a default inference to the success of an attempt to dig all the way through. On a normality-based approach, then, the claims in (16) are false: their target outcomes fail in all normal projections based on (rational subsets of) reference-time facts.

Unfortunately, however, the same prediction of falsity applies to progressives of tasks which are possible, but extremely unlikely to succeed. (17) illustrates: even though there is no minefield-aware perspective which licenses a default inference from an attempted to a successful crossing, (17) can be true in a situation where Trapper John has taken some steps into the danger zone.[10]

(17) Trapper John was crossing a live minefield.

A similar example, due to Bonomi (1997), shows that the progressive of a telic predicate P can be true even where non-culmination (the failure of C_P) is itself the most likely outcome of any good-faith attempt to do P. The immediate context for (18) (and, presumably, the announcer's domain-specific expertise) preclude a default inference to the success of any single boat in circumnavigating the globe; (18a) is nevertheless both acceptable and true.[11]

[10] Note that the acceptability of (17) does not rely on an "ignorant" perspective: it remains valid in a context where all participants, Trapper John included, are apprised of the danger (M*A*S*H, Season 2, Ep.6).

[11] As Bonomi (1997) points out, Asher's truth conditions for (18a) require that the following is a valid default; the spokesman's first and second sentences should be contradictory, but they appear perfectly compatible.

(1) For any boat x in the competition which is circumnavigating the globe, x typically circumnavigates the globe, eventually

(18) **The sailing scenario.** Suppose that an international sailing association organizes a competition whose goal is the circumnavigation of the globe. After thorough selection, 100 boats are admitted, and all sail on the scheduled date. Some days later, a spokesman says:

a. 100 boats are circumnavigating the globe. Most of them will fail.

From one point of view, the difference between unacceptable progressives of impossible events and potentially-valid progressives of merely **unlikely events** has to do with the set of possible outcomes of an attempt to do P. Both Dowty and Asher treat the progressive as a type of *universal* modal, requiring successful culmination across the full set of relevant alternatives. Impossible task progressives fail this requirement because *none* of the normal/inertial alternatives (whether generated globally or from a restricted perspective) realize C_P, but the same need not be true of unlikely event progressives. We might then thread the needle between these two types of data by weakening the modal force of an intensional progressive operator: the progressive of some predicate P would then be true of a reference time state s just in case some admissable perspective on s has at least one normal continuation that realizes C_P.

As it turns out, even an existential intensional progressive makes the wrong predictions for telic predicates in contexts where culmination is *locally* **out of reach**.[12] The target task in (19), for example, is certainly not categorically impossible, but successful culmination is precluded in the local context by (potentially mutable) properties of the protagonist Benny.[13]

(19) **The undertrained runner** (modified from Varasdi 2014, pp.192–193). Benny is an amateur but ambitious runner who decides to run an ultramarathon. He will not in fact be able to run the whole distance, because very few can, and he does not have the necessary physical qualities, having trained insufficiently to build up his endurance to the required degree. Nevertheless, he registers for the race, appears on the given date, gets his race number, and sets off with the other runners. The first couple of kilometers go well, but then he begins to lose strength, and at about half the distance he collapses because of exhaustion.

a. *Target:* Benny was running an ultramarathon (when he collapsed).

From an 'objective' perspective—say, that of a speaker who has learned Benny's name, but does not know him personally—the problem with (19a) is the same as the problem of (18). While we may be able to define an admissable perspective on (19) which excludes details about Benny's training and endurance, it still seems that the default inference required for the truth of (19a) (namely, that completing an ultramarathon is the *typical* result of running in one) should be precluded by basic information about the race: most attempts to run an ultramarathon do not conclude successfully. The problem goes beyond this, however: (19a) seems to be acceptable and true even when we

[12]Dowty (1977) considers an existential analysis but sets it aside based on (1) (Dowty 1979, citing R. Thomason):

(1) **The fair coin scenario.** Assume a fair coin has been tossed, and at reference time is still rising.

a. ??The coin is coming up heads.

b. ??The coin is coming up tails.

Since a fair coin will come up heads in half of the inertial futures, and tails in the other half, (40a)-(40b) should both be true—against intuition—if the progressive introduces existential modal quantification.

[13]Szabó (2004) provides the example in (1), illustrating much the same point as (19).

(1) As the architect was building the cathedral he knew that, although he would be building it for another year or so, he couldn't possibly complete it.

assume it to come from a 'subjective' speaker (say, a close personal friend), who is well aware of both Benny's physical limitations and the ensuing inevitability of his failure.

The judgments for (19a) in an out of reach context simply cannot be explained on an inertia- or normality-based approach to the progressive, regardless of modal force: *all* of the normal futures projected from (a subjective view of) the reference time situation must be ones in which Benny does not complete the race. If the intensional approach is correct, out of reach progressives should pattern with impossible event progressives. That this is evidently not the case shows clearly that there is some distinction between impossible and out-of-reach tasks to which the progressive is sensitive. The received Type I approaches fail to account for this distinction.

The preceding discussion suggests that an account based on completion requirements across all or even some 'normal' worlds is not the right approach for telic progressives. Indeed, Szabó (2008) takes data like (19) as evidence that the truth of telic progressives is not grounded in the modal accessibility of culmination: "events that cannot possibly culminate can nevertheless be truly said to be in progress, as long as we are allowed to abstract from those features that made their completions impossible" (Szabó 2004, pp.38–39, see also Varasdi 2014). From this viewpoint, the right approach must be a partitive one, on which the truth of telic progressives hinges on what is actually going on at reference time, rather than the 'normal' or expected consequences of these events. This is consistent with what Landman (1992, p.13) calls the 'part-of proposal'.

(20) **The part-of proposal.** *Mary is crossing the street* is true iff some actual event realizes sufficiently much of the type of events of Mary's crossing the street.

The load-bearing element in (20) is the phrase "sufficiently much". A partitive approach to the imperfective paradox must make reference to some set of criteria according to which an in-progress but non-culminated eventuality 'counts' as (part of) a P-event. This set of features must, intuitively, have a connection to the culmination condition of the underlying predicate P. However, if the partitive approach is to fare better than the intensional progressive on 'out-of-reach' data like (19), the connection cannot be formalized purely in terms of a local (modal) expectation of culmination. Put another way, any relationship between reference time activities and P's culmination condition must be realized at the level of the abstract *event type* denoted by P, rather than in terms of a reference time token. The challenge thus lies in establishing precisely what it means for some set of (reference time) facts to realize *sufficiently much* of a P-type event.

2.3 Parts and measurement

The best-known extensional (Type P) account of the imperfective paradox is due to Parsons (1989, 1990). He argues that in making truth conditional assessments of inflected telic predicates, it is always and only "the present activities that are the whole story," rather than "the idea of what would be the case [. . .] if present activities were to go on uninterrupted" (1989, pp.221-222). In the progressive context, this approach necessitates letting go of the culmination assumption, in order to allow a non-culminated eventuality to qualify as a valid instantiation of a telic predicate P.

Parsons shifts the responsibility for culmination entailments (where they arise, as in 7a) from the denotation of an uninflected telic predicate to the semantic contribution made by a modifying aspectual operator. In his framework, progressives and nonprogressives (perfectives) do not differ in their intensionality, but in which features (roughly, how much) of a P-type eventuality they require reference-time events to realize.[14] Example (21) schematizes Parsons's proposals: both perfective (21a) and progressive (21b) require that a reference time event e satisfies some core characterization provided by the uninflected predicate, but differ in whether or not e must *culminate* at reference time, or can merely *hold*. Crucially, while the relation $\mathsf{Cul}(e, t)$—imposed by a perfective or simple past—is true of an event e at time t just in case e culminates (in some relevant sense) at time t, $\mathsf{Hold}(e, t)$—imposed by the progressive in place of Cul—neither implies nor predicts culmination, requiring only that "e is an event which is in development at t" (1989, p.220).[15]

(21) a. Mahler wrote a symphony. $\sim$ PST(PFV($\sqrt{}$*Mahler write a symphony*))
$\exists e[\mathsf{writing}(e) \wedge \mathsf{Agent}(e, \mathsf{m}) \wedge \mathsf{Theme}(e, \text{a symphony}) \wedge \mathsf{Cul}(e, t\{\prec_i \mathsf{now}\})]$

b. Mahler was writing a symphony $\sim$ PST(PROG($\sqrt{}$*Mahler write a symphony*)).
$\exists e[\mathsf{writing}(e) \wedge \mathsf{Agent}(e, \mathsf{m}) \wedge \mathsf{Theme}(e, \text{a symphony}) \wedge \mathsf{Hold}(e, t\{\prec_i \mathsf{now}\})]$

To the extent that this approach obviates the imperfective paradox, it has a clear conceptual appeal. From a technical perspective, however, Parsons's explication of Hold leaves something to be desired: namely, what it means for an event in the denotation of a telic predicate to be "in development". A partial answer emerges in the logical consequences of combining a Hold requirement with the first three conjuncts of (21b), which come from the underlying predicate ($\sqrt{}$*Mahler write a symphony*) itself. The specification of a semantic Theme in a representation like (21b) forces the reference time event e to involve some object belonging to the denotation of the nominal predicate *symphony* (see also Landman, 1992). In conjunction with Hold, however (again, as opposed to Cul), this thematic object can at best be a *partial realization* of a symphony. Parsons treats this result as a positive one: he accepts that truth conditions like those in (21b) commit him to the existence of a symphonic object, but suggests that where the reference time event e holds (but does not culminate), the object is simply incomplete.

For Parsons, the question of partitivity (partial realization) in telic predicates reduces to partitivity in the nominal domain. This analogy, based on parallels between mass nouns and atelic predicates versus count nouns and telic predicates, follows Bach (1986). Parsons argues that incomplete objects, like an unfinished piece of music, can still be considered instantiations of their intended wholes. He suggests that non-culminated events can similarly instantiate telic predicates, as their (in)completeness can be measured by the (in)completeness of the objects involved, particularly through an incremental theme. However, this link between event and nominal partitivity relies on the existence of a tangible correlate of development and fails with accomplishment predicates that lack appropriate thematic arguments (in particular, *incremental themes*; Dowty 1991; Rothstein 2000). Landman (1992) contends that certain 'sudden' creation processes, such as the progressive *God was creating a unicorn*, involve necessary developmental stages that partially

[14] In the standard semantic approach to distinguishing the (im)perfective aspects (see 8, 10), both operators compose with a predicate of events. Parsons adheres to this view. An alternative, metaphysical approach from the philosophical literature instead associates each aspect with a distinct type of ontological entity. Specifically, sentences with the imperfective aspect assert the occurrence of a 'process,' while those with the perfective aspect assert that an 'event' has occurred (Mourelatos 1978; Stout 1997; Steward 2013; Baratella 2023.) Similar to the Type P approach, this metaphysical perspective avoids the imperfective paradox, as processes do not entail culmination; culmination is an event. According to Steward and Baratella, processes expressed by the progressive have an intensional aspect, involving possible worlds. However—unlike the account we develop here—these proposals do not address how to verify the instantiation of either a partial or complete process. These accounts also fail to explain why telic perfectives entail their progressive counterparts, as illustrated in (7). See Section 4.3 for further discussion, and the appendix for an elaboration on the notion of process in the different approaches

[15] Hold therefore picks out processes, regardless of the aspectual class of the underlying predicate; see the appendix.

instantiate the telic predicate, even in the absence of an (incomplete) physical object.

Judgments about divine creation may seem obscure, but Landman's argument can be reconstructed with a more mundane example, such as baking a cake. This type of creation process is not sudden, but is crucially non-homogeneous: while a completed cake is a fully baked product, and there are various tangible correlates of development, no object that can be described as even an incomplete cake comes into being until the events in question are well underway. For instance, an event in which some agent mixes together appropriate ingredients intuitively counts as a cake-baking in progress, even though the contents of the mixing bowl cannot at this point be called a cake. This issue is even more apparent at earlier stages of cake-baking: *Emanuel is baking a cake* is true when he has merely collected the right ingredients and preheated the oven, but there is at this point no single object which could constitute the thematic cake required by Parsons.[16]

We believe that there is something important in common between telic predicates of sudden or 'non-homogeneous' creation and those with incremental themes. In each case, the truth of the progressive hinges on some measure of progress through a sequence of steps which lead in principle to the right kind of culmination. For predicates with incremental themes, it just so happens that steps along the pathway towards culmination are accompanied by (monotonic) incremental changes in the physical extent of an object, but this effect is secondary: it is not the presence of the physical correlate that validates the progressive claim. The cake-baking example underscores this point: even in the absence of a(n incomplete) thematic object, Emanuel's actions represent progress toward cake-baking. More precisely, Emanuel has taken steps which are necessary for cake-baking—they correspond, very nearly literally, to the steps one might find in a cake-baking recipe. This correspondence allows us to evaluate both whether he is indeed baking a cake as well as how far along he is in the process (*how much* of a cake-baking event he has realized).

The perspective outlined here aligns with Parsons in certain respects. First, it focuses on actual events (rather than projected future developments) as the entities that can verify or falsify a telic progressive claim. We also agree with Parsons that event partitivity is conceptually parallel to nominal partitivity, and that a treatment of the former must incorporate a framework for measured comparison. Crucially, a partial realization of some event type (like a partial object) is partial becuase of the relationship it has to a conceptual whole: it must contain at least some of the components that can be identified in the whole.[17] However, we do not believe that event partitivity can be assessed using the same measurement framework as object partitivity: the former does not reduce to the latter. Instead, the conceptual whole in the event domain must correspond to a well-defined 'recipe' or strategy for achieving culmination. The underlying intuition is this: in order to decide whether Emanuel is baking a cake, one must first know *how a cake is baked.* Only then can one assess whether Emanuel's actions correspond, step by step, to the components of the conceptual whole that constitutes cake-baking.

Insofar as it requires a framework for measuring progress through a culmination 'recipe', partitivity in the domain of complex (telic) eventualities is dependent on the specification of a culmination condition. Our task, then, is to find some way of formalizing the relationship between process steps (line items in a recipe, to continue the metaphor) and culmination in such a way that

[16] This discussion is related to the issue of *futurate* uses of the progressive (Copley, 2009), which appear to target preparatory events that lead (causally) to the initiation and development of the input predicate. We leave a more careful exploration of this connection for future work.

[17] Put another way, just as identifying an object as *part of* a house involves comparing its components to those of a complete house, identifying an actual event as *part of* a house-building involves comparing its components (subevents or stages) to a complete or *culminated* version of the same kind of thing.

measurement and comparison can be achieved on the basis of "present activities" alone. The link between a culmination procedure and the successful culmination of a telic predicate must therefore be established within the denotation of an uninflected predicate itself, and cannot rely on local (reference time) modal projections about the future. We will argue that causal information lies at the heart of the process-culmination relationship.

2.4 Taking stock: interim summary and desiderata

The preceding sections established desiderata for a satisfactory account of telic non-culmination in the progressive context. We review these points here, using them to motivate a particular descriptive generalization as the target of our analysis, and to clarify the analytical challenges which a causal-modelling approach is intended to address.

Section 2.2 introduced three key classes of data: (telic) progressives of *impossible events* (IEs), progressives of *unlikely* or *unexpected events* (UEs), and progressives in *out of reach contexts* (ORCs). Since intensional Type I approaches require reference time facts to predict successful culmination across an appropriately constrained and locally-projected set of modal alternatives, these accounts predict falsity across the board. This is compatible with empirical judgments for IE progressives (see 16a-16b), but it does not reflect the empirical data in the case of UE and ORC progressives, which can be both acceptable and true, as shown in Table 1.

Data type	Examples	Empirically	Type I prediction
IEs	(16a) Mary is swimming across the Atlantic Ocean (16b) The children are digging a hole to China	✗	false
UEs	(17) Trapper John was crossing a minefield (18) 100 boats are circumnavigating the globe	✓	false
ORCs	(19) Benny was running an ultramarathon (1) The architect was building the cathedral	✓	false

Table 1: Intensional progressive (Type I) predictions for IE, UE, and ORC progressives

Based on the judgments reported above for UE and ORC progressives (see also data from from Bonomi 1997; Szabó 2004, 2008; Varasdi 2014), we argue that the central tenet of Type I approaches—i.e., that the truth of telic progressives relies on locally-projected alternatives—cannot provide an empirically adequate account. Appropriate truth and acceptability conditions must instead be based on something that the input predicates for UE progressives and those for ORC progressives have in common with one another, but fail to share with predicates of IEs. Moreover, since UE and ORC progressives do not share the same (local) intensional relationship to culmination, the pattern of judgments reported in Table 1 suggests that, where a telic progressive is acceptable, its truth-conditional assessment must be based on information that pertains only to reference time facts, and not to their expected (projected or inferred 'default') developments.

This suggests that a partitive Type P approach might be preferable; we therefore examined Parsons's (1989) extensional account in Section 2.3. While we agree with the spirit of Parsons's analysis—in particular, that a telic progressive requires the actual (evaluation world) instantiation of some eventuality constituting a well-defined *part* of a complete (culminated) eventuality as denoted by a telic predicate P—we argued that the requisite conception of *parthood* cannot be based on a tangible or physically measurable correlate of the progress of a P-eventuality, since not all telic predicates invoke an appropriate physical dimension. We suggested, instead, that the appropriate basis to use in comparing reference time facts to a complete (culminated) P-eventuality is that of a procedure or 'recipe' for culmination: more precisely, that an evaluation world event can qualify as a (potential) *part* of a P-eventuality just in case it realizes some portion (set of steps) that belong to a recognized *procedure* for realizing P's culmination condition. With respect to the Table 1 data, this means that the sense in which Trapper John is crossing a minefield, the competitor boats are circumnavigating the globe, Benny is running an ultramarathon, or the architect is building a cathedral at their respective reference times is that in each case, the (relevant) set of reference time facts corresponds to some part of a strategy—a culmination procedure—which one might adopt in setting out to complete an event of the right type.

The concept of a **culmination procedure**—to be given formal content in the next section—not only explains the acceptability of UE and ORC progressives in the contexts referenced in Table 1, but moreover serves to distinguish them from the unacceptable IE progressives in (16a)-(16b). We wish to emphasize that impossible tasks are just that—impossible—because *there is no way to complete them*: not only is there no historical record of successful completion, but there is also no imaginable method for achieving completion (within the bounds of common sense). The availability of a world-historical culmination procedure clearly divides IE from UE and ORC progressives: while participants in UE and ORC tasks can make progress towards culmination by matching their activities to some recognized culmination procedure, there is no such process or 'recipe' to follow in the IE case. Consequently, IE predicates provide no basis on which to perform a partitive assessment of reference-time event.

We propose the following descriptive generalization as our analytical target:

(22) **Descriptive truth and felicity conditions for telic progressives.** Let P be an accomplishment predicate with culmination condition C_P. Then:

a. $\mathsf{PROG}(P)$ is felicitous (subject to truth-conditional evaluation) just in case P is associated with at least one culmination procedure (roughly, a 'recipe' for realizing (C_P)).[18]

b. Where felicitous, $\mathsf{PROG}(P)$ is true just in case two conditions are met: First, the reference time situation s can be partially 'matched' to a culmination procedure associated with P (i.e., where s maps onto a snapshot or cross-section of a partial P-eventuality, as defined by partial realization of a culmination procedure); second, reference time facts do not already settle that C_P is false (cf. Varasdi's (2014) notion of "sustain".)

Per (22), the truth and felicity of $\mathsf{PROG}(P)$ does not depend on locally-projected developments of the reference time situation, but it does rely on a formal link between what is true at reference time and the structure of an abstract P-eventuality, which in turn is defined in reference to culmination. From our perspective, (22) combines the key insight of Type I approaches—again, that an in-progress P-eventuality must be one which aims at culmination in an inertial or closed-world context—while making good on the mainstay of a Type P approach: that reference time activities matter more than future developments in determining whether or not a P-eventuality is *in progress*.

[18]There are certain similarities between this suggestion and the notion of a process eventuality invoked by Lascarides (1991); for her, the content of such an eventuality is simply supplied by world knowledge, while we ultimately propose that world knowledge about such processes is given set-theoretic structure by means of a formal causal model.

The success of the above analysis will, of course, rest on how the key notion of a *culmination procedure* is cashed out. The aim of Section 3 is to provide a formal framework for the proposal in (22); we therefore conclude this section by discussing the challenges for an adequate treatment of culmination procedures and their relationship to the lexical representation of a telic predicate.

We have already suggested that causal information is central for a theory of event partivity; since event partivity, in turn, is based on adherence to a *culmination procedure* (as per 22), causal information must also be central to the key procedure concept. In centering causal information this way, we are not advocating for the *causative* analysis of accomplishment predicates that follows from a traditional decompositional treatment such as the one sketched in (2) (see Section 1). Our main objection to the causative treatment is worth reiterating: an actual (singular, or token) causal relationship between a partial (in-progress) P eventuality and the culmination condition of P can only be evaluated in contexts where both the causing action and the culmination condition are realized. Since telic progressives can be true where culmination fails, events satisfying a telic progressive need not verify an actual causal relationship in which the relevant culmination occurs as the caused eventuality: indeed, to the extent that ORC progressives can be both acceptable and true, events satisfying a telic progressive in many contexts *cannot* verify such a relationship (in the evaluation world or any accessible modal alternative).

Nevertheless, it seems to us that there are contexts in which a felicitous and truthful use of a telic predicate can and should verify relations of actual causation: these include contexts (e.g., perfective uses) in which the realization of the culmination condition associated with telic predicate P is entailed. Any context in which it is true that *Trapper John crossed a minefield*, *Benny ran an ultramarathon*, or *The architect built a cathedral* must be a context in which the sentential subject was actually involved in some activity which *brought* the relevant culmination condition *about*. Cases where this relationship does not hold—i.e., in which the sentential subject does not participate in any relevant activity, the culmination condition goes unrealized, or in which no relation of actual causation obtains between the subject's activities and successful culmination—straightforwardly fail to verify telic perfectives. Thus, assuming that any 'process'-only part of a culminated telic eventuality should qualify as an in progress P-eventuality, we arrive at a conclusion made explicit in Szabó (2004): the truth of a telic perfective requires not only the prior truth of its progressive correlate, but also the truth of an actual causal relationship between the progressive eventuality and the actual culmination of the underlying predicate.[19]

Here is the balancing act. The information supplied by a telic predicate should permit verification of some set of actual causal relationships—between a developing (process) stage of a P-eventuality and its actual culmination, or between developmental stages of an incomplete P-eventuality (see Szabó 2004, and fn. 19)—*modulo* how much of a P-eventuality is actually instantiated. On our view, it is the reality of these relationships—i.e., the sense in which each stage in the development of a P-eventuality *leads to* its further development, up to the point of culmination—which gives rise to the intuitions about 'inertial' continuations which motivate Type I analyses. At the same time, however, making the truth of a telic progressive contingent on its actual causal relationship to culmination excludes a satisfactory treatment of ORC data. Thus, whatever qualifies a course of events as a valid (partial) instantiation of a telic predicate—or, in the terms in (22), permits a match between a reference time situation and some stage in a culmination procedure for P—must be verifiable independently of actual causation. It must, moreover, be conceptually prior to the verification process for actual causation, since it is *in view of* an event's status as a partial/in-progress P-eventuality that it is able to verify actual causal relations in the first place.

We suggest that accomplishment predicates are not defined in terms of instances of actual causation, and, accordingly, that their inflected uses do not correspond directly to causative claims. Nevertheless, the truthful and felicitous use of an accomplishment predicate can license actual causal claims: this is why a culminated use of a telic predicate entails an actual causal link between some (prior) event involving the sentential subject and the realized culmination. In Section 3, we argue that singular (actual) causal claims are licensed by **type-level causal information**, some relevant body of knowledge that encompasses generalizations (gleaned from observation and/or experience) about causal relationships in the world. Thus, if (particular uses of) accomplishment predicates can license singular causal claims, it follows that these predicates introduce type-level causal information, pertaining to the conditions under which the relevant sort of culmination is typically realized. An uninflected telic predicate must then contain idiosyncratic components related to both culmination and the process which leads to it. Below, we argue that type-causal information is naturally supplied by a **causal model**: we begin with an overview of causal models

[19] On the basis of telic progressives' acceptability in out of reach contexts like (1), Szabó (2004, p.40) arrives at a similar conclusion to Section 2.2: that Type I approaches to the imperfective paradox are doomed. Rather than attempting to analyze telic progressives *in terms of* the truth conditions of their perfective correlates, Szabó suggests a 'reverse' analysis, on which telic perfectives are treated as entailing strictly more than their progressive correlates (see 7). His proposal is sketched below: it makes clear not only the requirement that a relation of actual causation should obtain between some in-progress P-eventuality and the result state ($\textsc{tel}(P)$) associated with P, but moreover that a process for realizing $\textsc{tel}(P)$ must be subject to a complex set of internal causal relationships, each of which should be actually verifiable in a context in which a complete P-eventuality is actually instantiated.

(1) **The 'reverse' analysis** (adapted from Szabó, 2004, p.47). If P is telic and admits the progressive then $\textsc{pfv}(P)$ is true iff:

a. $\mathsf{PROG}(P)$ is true of some event e

b. $\textsc{tel}(P)$ is true of some state s

c. e causes s

d. If e causes e' and e' causes s, then $\mathsf{PROG}(P)$ is true of e'

We are interested in the truth conditions of telic progressives, which Szabó does not discuss. However, although we deliberately avoid committing to the claim that a telic perfective's truth is *constituted* by the actual causal relations in (1), we believe that our account is compatible with (1), insofar as the conditions above will follow from the truth of a telic perfective (at least for those perfective operators which license culmination entailments; see Section 4).

and their relevance for singular causal (causative) claims, ultimately proposing a formal treatment of telic predicates within the *structural equation* (SEM; Pearl 2000) framework.

3 Causal models for event types

When we engage in 'practical' (language-independent) causal reasoning—for planning, experimentation, and so on—we conceptualize caused events as resulting from the joint contribution of multiple prior conditions. For instance, whether or not a match ignites certainly depends on whether or not it was struck on a suitable surface, but also on a range of other factors, including (but not limited to) whether or not it was dry at the time of striking, and whether or not oxygen was present: a realistic understanding of the ignition process includes the knowledge that if the match is wet or there is no oxygen, no amount of striking will ever succeed. Causation thus emerges as a multifaceted, many-to-one relation, in which effects are linked to sets of (potentially interconnected) causes. We refer to any structured representation of these complex relationships as a **causal model.**

The complex structure of a cognitively-plausible causal model stands in contrast with standard linguistic representations of causation, which express apparently binary relationships between single causes and effects (see 23a-23c). The prevalence of such one-to-one descriptions motivates classical decompositional approaches (see, e.g., 2/3 in §1), on which causal expressions are uniformly analyzed as encoding a single binary dependence relation—CAUSE—between two events. When considered in relation to causal models, however, CAUSE-decompositions face a number of challenges. On the metaphysical side, reducing any kind of causal dependence to a single relation raises the longstanding issue of *causal selection*: it is clear that (23a) provides a better causal description of a normal match ignition than does (23b), but it is not at all clear what principles privilege the selection of striking over oxygen if a realistic conceptual model for match ignition places both conditions as causally prior to and jointly responsible for an ignition event (Baglini and Bar-Asher Siegal, 2025, a.o.). On the linguistic side, uniform CAUSE-decomposition is difficult to reconcile with the sheer diversity of contrastive causal expressions: given an account of causal selection which explains the difference between (23a) and (23b), how can we account for the acceptability of (23c) on a theory which forces *cause* and *enable* to introduce the same core causal relation?

(23) a. Striking the match caused it to ignite.
b. ??The presence of oxygen caused the match to ignite.
c. The presence of oxygen enabled the match to ignite.

One solution is to set aside the notion of a uniform lexical atom like CAUSE. If the 'practical' concept of causality reflects cognitive reality, causal models can be treated as potential inputs to linguistic description, becoming objects which are subject to reference, update, and manipulation by the usual linguistic means. Call this the **modeling perspective** on linguistic causation: the upshot is that descriptions of causal situations—like descriptions of any other sort of situation—need not capture the entire causal picture. Instead, causal language references particular facets of a contextually-relevant model. From this perspective, causal descriptions are unified by their shared reference to causal models (i.e., to a particular kind of contextual parameter), but there is no reason that distinct expressions should be uniform in their selection criteria: *cause* and *enable* in (23), for instance, share reference to the match-ignition model, but differentially constrain the role that their subject-cause(r)s take within a set of causes acting together to produce the specified result. This approach is adopted by Nadathur and Lauer (2020) in the comparative semantic analysis of periphrastic causative verbs, and by Baglini and Bar-Asher Siegal (2025) in accounting for 'directness' inferences associated with lexical vs. productive (periphrastic or morphological) causative descriptions (see also Bar-Asher Siegal et al. (2021); Bar-Asher Siegal (2025)).

The causal modeling perspective will only be explanatory if it is paired with a representational framework which captures the many-to-one nature of practical causal knowledge while providing a structure in which differences between elements of a causing set can be clearly articulated (see also Hobbs 2005). We use *structural equation models* (SEMs) to provide a formal implementation for linguistically-relevant causal models. Developed by Pearl (2000) and colleagues (Spirtes et al., 1993; Steyvers et al., 2003; Halpern and Pearl, 2005; Halpern, 2015, a.o.), SEMs are regularly used in the study of causal inference and explanation, decision making, and related phenomena across an array of disciplines in the cognitive and computational sciences: this makes them highly plausible candidates for the reification of practical causal knowledge. Our choice of framework is also motivated by recent work which has fruitfully employed SEMs in the study of a range of linguistic phenomena, including the semantic analysis of counterfactual conditionals (Schulz 2011; Henderson 2010; Bjorndahl and Snider 2015; Ciardelli et al. 2018) and the lexical analysis of causative constructions (see references above), as well as expressions that are less obviously causal in nature (such as implicative and abilitative predicates; Baglini and Francez 2016; Alonso-Ovalle and Hsieh 2017; Copley 2021; Nadathur 2023a,b).

3.1 Causal claims and causal models: token vs. type causality

Before introducing the formal machinery of SEMs, we want to highlight some of the consequences of adopting the causal modeling perspective on linguistic causation. At its core, this perspective takes a causal model to be a repository of causal information which may be relevant for non-linguistic purposes (e.g., deciding how to act in a given situation) as well as for linguistic ones (e.g., deciding how to describe the situation). Crucially, a model is a language-independent cognitive object which encodes a set of generalizations reflecting what is known or believed about how causality works in a given system. These generalizations (model-internal relationships) may produce causally-driven expectations in particular contexts, but they are not *about* any particular context.

In contrast, episodic causal claims like those in (23) and (24) are explicitly focused on singular contexts: such statements adhere to the general pattern of non-modal declarative claims, appearing to describe complex but concrete scenarios (i.e., *what happened*) at specific spatiotemporal locations. The apparent truth conditions of a periphrastic causative claim like (24) are represented schematically below: (24) requires the realization of a causing (vase-dropping) eventuality, the realization of a result (vase-breaking) eventuality, and that the former led to the latter.

(24) Ria dropping the vase caused it to break.
  a. Ria dropped the vase. (event c)
  b. The vase broke. (event e)
  c. Event c (is part of what) *brought about* event e.

The difficulty with (claims of) singular causation lies in dependency conditions like (24c) (cf. Hume 1748). While (24a) and (24b) report on concrete, directly observable facts, (24c) does not. We cannot, of course, discard a condition like (24c) as truth-conditionally relevant for (24), since it is intuitively clear that "unlinked" occurrences of c and e are insufficient for the truth of (24), even if both events take place within reference time and in the correct temporal order ($e \not\prec c$). To the extent that the semantic interpretation system relies on the verification of particular atomic predications, the schematic truth conditions in (24) necessitate the existence of a reference object for verifying the causal relationship in (24c): this object must be something which goes beyond observable events at the referential here-and-now. From the causal modeling perspective, this is precisely what a (salient, contextually-relevant) causal model supplies: an abstract but structured object against which causal predications can be checked.[20]

Causal models thus provide licensing conditions for causal claims: the truth and appropriateness of (24) depends not only on the occurrence of events c and e, but also on the availability of a model which encodes a causal relationship (i.e., a generalization) linking occurrences of events of the c type to events of the e type.[21] On the modeling perspective, a suitable formal framework for describing the structure of a causal model should allow such a model to act as a bridge between world knowledge (of a causal nature) and the familiar objects of formal semantics, representing causal knowledge in a format which allows us to express precise (set- and/or model-theoretic) conditions under which a relation corresponding to (24c) is verified.

Causal models are thus not "about" causal statements, but are instead "about" what individuals know of the world. Although we hold that singular causal claims like (24) rely on causal models, we also do not intend to claim that these statements are *about* causal models. Singular causal claims are, as they seem to be, about particular situations and events in the world: it is simply that they are made *in view* of situationally-relevant causal knowledge. The distinction between causal models and singular claims can be likened conceptually to the distinction between **type-** and **token-level statements**. Type-level claims are recognizable as generic statements, expressing overarching patterns or 'laws' in the world: causal generics like "Smoking causes cancer," which seem to express relations between categories of events, illustrate this concept in the causal domain. Token-level claims are necessarily episodic, highlighting individual occurrences at particular places and times. Singular (e.g., past-tense) causal claims are of this sort, emphasizing a localized cause-and-effect link but failing to describe (or ascribe) any general links between categories of events.

If we take a scientific (empirically-driven) point of view, of course, singular causal claims are necessarily understood as descriptions of the sorts of real-world situations which must be regularly or repeatedly observed in order to formulate causal generalizations, and thus to license acceptable type-level causal claims. The flow of comprehension, then—i.e., the development of causal knowledge—must run from the specific to the general, or from token instances of causation to type-level models: from this perspective, a model is simply a reification of causal regularities drawn from robust observation of specific causal instances (Hausman, 1998, 2005). A causal model is thus ostensibly constructed on the basis of observation, experimentation, and experience; the relevant statistical and qualitative correlates of model-construction are a matter of research in cognitive science and philosophy (see Woodward 2003; Hitchcock 2020), but will not concern us here.

It is the reverse perspective—moving from the general to the specific—which is our chief interest. Once a causal model is established (has become cognitively available), the causal generalizations

[20]This is not the only solution to the problem posed by causal 'unobservability'. The alternative is a reductionist approach, on which the causal relationship in (24) is decomposed into modal (alethic) statements: this approach is prominent in the philosophical literature on causation (see, e.g. Mackie, 1965; Lewis, 1973a; Sosa and Tooley, 1993), and (to the extent that it coheres with the CAUSE-decomposition approach) is often presupposed by linguistic work (Dowty, 1979). Bar-Asher Siegal and Boneh (2020) provide a discussion of the relevant philosophical literature and its points of connection (and contention) with the linguistic literature and data (see also Copley and Wolff 2014).

[21]A similar idea can be found in Davidson (1967),(in turn following Hume 1748) who regards causation or causal laws as a set of nomological generalizations: i.e., statements of lawful regularity that may be inferred through experience.

which it encodes correspond to a set of *nomological* relations between certain types of objects (conditions, properties, punctual/point events), by virtue of which token-causal claims may be verified. The truth conditions of an individual instance of causation—in particular, a condition analogous to (24c) for (24)—are formulated in terms of these type-causal relations (see also Tooley, 1987; Hoover, 2001). As a result, even though the causal relation itself is not directly observable, causality can be attributed to specific states of affairs based on our understanding of type-level relations in the world. For instance, (24) reports on the realization of a vase-dropping action by Ria, a state transition involving the vase, and indicates that the actual occurrences of these events conform, in their relative spatiotemporal locations, to a nomological relation between events of the dropping- and state-transition types in a causal model. In order for her to felicitously utter (24), this causal model must be (a) available to the speaker and (b) presumed by her to be shared with her audience (cf. Baglini and Bar-Asher Siegal 2025; Bar-Asher Siegal 2025). This causal model encodes knowledge about types—specifically, general principles governing the culmination of processes. As noted above, type-level knowledge is crucial for evaluating, at an intermediate stage of a process, whether it is on course to reach its intended culmination. This conceptual framework will play a central role in our analysis of the semantics of telic predicates. With this in place, we now return to our primary concern: how such causal and temporal structures inform the interpretation of telicity.

3.2 The formal framework: structural equation models for causation

Practical causal reasoning requires three things: an actual context or situation, a set of salient or relevant causal laws (a model), and a mechanism for applying the latter to the former. We use the Lifschitz circuit context in (25) to illustrate: (25a) states the relevant causal laws, and (25b) provides a context in which these laws are presumed to be in effect.

(25) **The Lifschitz circuit.** Suppose we have a circuit with two switches and one light.

a. The light turns on just in case both switches are in the same position: either both up or both down.

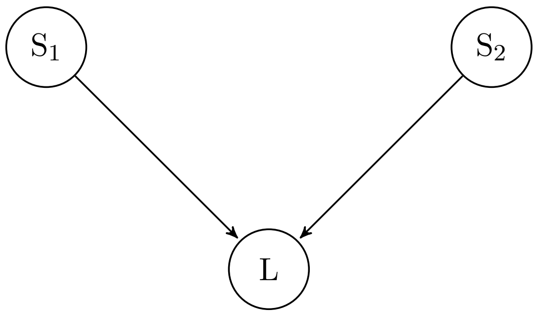
b. At the moment, switch 1 is down, switch 2 is up, and the light is off.

Given (25a), it is clear that (25b) represents a *causally normal* state of affairs: the state of the dependent element (the light) is as expected given the states of the independent elements (the two switches). The information in (25a) can also be used to make predictions about the world: namely, that flipping either of the switches will produce a change in the light. The availability of such predictions also allows us to establish links between flipping the switches and turning the light on or off, and to make specific causal claims about the relations between these events.

The information in (25a) expresses the type-level causal relations which should be represented in a model for the circuit. Figure 1 represents these relations more formally. The graph in 1a conveys that (the state of) the light L causally depends on (the states of) the switches S_1 and S_2. The three-valued truth table in 1b specifies the nature of the dependencies, indicating that the light comes on (L takes on value 1) just in case S_1 and S_2 have the same value (both 0 or 1; let 0 stand for ‘off’ and 1 for ‘on’); the third truth value, u, represents that a node’s status is unknown.[22] The

[22]The use of a three-valued epistemic system (cf. Schulz 2011) is consistent with our aim of modeling speakers’ causal knowledge within the SEM framework. Epistemically, causal knowledge concerns the inferences that speakers are able to draw about causal relations, even in the absence of complete information about all relevant conditions.

causal normality of the situation in (25b) is reflected in a correspondence between the actual values for S_1 (0), S_2 (1), and L (0) and the sixth line of Figure 1b; the causal predictions mentioned above can similarly be read from 1b by moving to a line in which one or the other of the switches changes its value, and noting that the corresponding value of L changes in each case.



(a)

$f_L =$

S_1	S_2	L
u	u	u
u	0	u
u	1	u
0	u	u
0	0	1
0	1	0
1	u	u
1	0	0
1	1	1

(b)

Figure 1: Graphical model for the Lifschitz circuit

Figure 1 is an example of a **structural equation model**. The skeleton of an arbitrary SEM comprises a *directed acyclic graph* $D = \langle V, A\rangle$ in which the set of vertices V corresponds to a finite set of (relevant) propositional variables, and the set of directed edges corresponds to a relation $A : V \times V$ which (in this context) represent an atomic notion of *causal relevance*. Vertices can be paired with *truth values* from the set $\{u, 0, 1\}$; a value of 0 or 1 identifies a propositional variable with a *proposition* in the familiar linguistic sense, while a u value indicates that the status of the vertex is *undetermined* (either unsettled or unknown). Given two vertices $X, Y \in V$, $\langle X, Y\rangle \in A$ indicates that X is an immediate causal ancestor of Y: more precisely, that the value of X influences the value of Y. We distinguish a set $I \subseteq V$ of *inner* or *dependent* variables: $I = \{X \in V | \exists Y \in V : \langle Y, X\rangle \in A\}$ (see also Schulz, 2011). In the model in Figure 1, the set V is given by $\{S_1, S_2, L\}$, the relation A by $\{\langle S_1, L\rangle, \langle S_2, L\rangle\}$, and the set I by $\{L\}$.

The graph indicates the existence of causal dependencies, but does not provide any information about their nature. This skeleton must therefore be associated with a (partial) function F that specifies *how* the value of a dependent variable is influenced by those of its immediate causal ancestors.[23] F maps each dependent vertex $X \in I$ to a pair $\langle Z_X, f_X\rangle$ where $Z_X = \{Z : Z \in V \wedge \langle Z, X\rangle \in A\}$ is the set of X's immediate causal ancestors and $f_X : \{u, 0, 1\}^{|Z_X|} \to \{u, 0, 1\}$ is a function from valuations of Z_X to valuations of X (cf. Schulz, 2011). As shown in Figure 1b, the *structural equation* f_X for a given dependent variable X can be expressed in the form of a truth table. It might, equivalently, be given as the (one-directional) logical form in (26); note that (26) should be taken to follow the strong Kleene logic for conjunction, in keeping with Figure 1b.[24] Given a *situation s*—a three-way assignment of truth values to V (see Definition 28b)—we can use the information in F to check for causal consistency (adherence to the model) as well as to predict

[23]Following Schulz (2011), linguistic applications of SEM typically assume that F should be *rooted* in the set $B =$ V I of *exogenous* (or *outer*) variables: this ensures that any path which traces backwards through the dependencies of a given variable will terminate in B, and prevents circular chains of causation. We will assume rootedness here.

[24]This differentiates our framework from that of Schulz, who uses a weak Kleene logic for causal consequences.

the causal consequences of any changes to s.

(26) Structural equation (f_L) for the Lifschitz example: $L := S_1 \wedge S_2$

The structure of an SEM allows us to define a variety of model-internal relationships. From the causal modeling perspective, the potential referents of causal expressions are distinct but well-defined types of structural dependency which may obtain between an inner variable X and one or more of its *causal ancestors* (informally, variables that are 'upstream' of X in the relevant model).

(27) **Causal ancestors.** Let $M_D = \langle D, F_D \rangle$ be a causal model over a set V of propositional variables. The set $\mathsf{Anc}(X)$ of *causal ancestors* for any $X \in V$ is given by $\{Y \in V | R_A(Y, X)\}$, where the relation R_A is the transitive closure of A (the set of directed edges of D).

Ultimately, we are interested in formalizing the notion of a **procedure** or 'recipe' for a given result: intuitively, this should constitute a causal pathway which picks out a set of *facts* (Definition 28a) that work together (i.e., are collectively *sufficient*) to produce some target effect, and which are independently required (*necessary*) for this effect (when considered as part of the pathway). The definitions in (28) provide a basis for formalizing these intuitions by allowing us to refer to variable-value pairs (facts), particular value assignments (situations), and the variables which are assigned 0 or 1 truth values in these assignments. A *fact* corresponds to the familiar linguistic notion of a proposition; note that a *situation* (Definition 28b) is uniquely identified its *kernel* (the set of facts it establishes; Definition 28b-iii), but not by its *domain* (Definition 28b-i).

(28) Let $M_D = \langle D, F_D \rangle$ be a causal model over a set V of propositional variables.

a. **Fact.** A *fact* is a pair $\langle X, x \rangle$ where $X \in V$ and $x \in \{0, 1\}$.

b. **Situation.** A *situation* $s : V \rightarrow \{0, 1, u\}$ is an assignment of truth values to V.

i. **Domain.** The *domain* $\mathsf{Dom}(s)$ of a situation s is the set of propositional variables to which s assigns a value other than u: $\mathsf{Dom}(s) = \{X \in V | \langle X, 1 \rangle \in s \vee \langle X, 0 \rangle \in s\}$.

ii. **Determination.** If $X \in \mathsf{Dom}(s)$, we say that X is *determined* by s.

iii. **Kernel.** The set of determinations made by a situation s is called its *kernel*: $\mathsf{Ker}(s) = \{\langle X, s(X) \rangle | X \in \mathsf{Dom}(s)\}$

iv. **Supersituation.** Given two situations, s and s', s' is a *supersituation* of situation s just in case $\mathsf{Ker}(s') \supseteq \mathsf{Ker}(s)$.

Model structure induces a notion of *causal consistency* (Definition 29). A situation s is consistent with respect to a model M just in case it is causally normal relative to M—i.e., if the values it assigns to a dependent variable X match the predictions of f_X, given the values s assigns to $\mathsf{Anc}(X)$. Thus, the situation $s = \{\langle S_1, 0 \rangle, \langle S_2, 1 \rangle, \langle L, 0 \rangle\}$ in (25b) is causally consistent with respect to the Lifschitz model in Figure 1, while the alternative situation $s' = \{\langle S_1, 0 \rangle, \langle S_2, 1 \rangle, \langle L, 0 \rangle\}$ is not (since it fails to conform to the structural equation f_L).

(29) **Causal consistency.** Let $M = \langle D, F_D \rangle$ be a causal model over a set V. Situation s is *causally consistent* iff, $\forall X \in \textsc{dom}(s) \cap I (\subseteq V)$, $s(X) = f_X(\overrightarrow{s(Z_X)})$

Causal consistency feeds *causal necessity* and *sufficiency* (Definition 30; see also Baglini and Francez 2016; Nadathur and Lauer 2020; Nadathur 2023b; Baglini and Bar-Asher Siegal 2025). These relations hold only between (collections of) facts, and thus can only be assessed in the context of a truth value assignment; moreover, since it must be the causal information in an SEM that determines whether necessity and sufficiency obtain, these definitions must make reference to (hypothetical) assignments—situations—that conform to the model. Per Definition 30a, causal necessity holds between two facts in a consistent situation just in case changing the value of the 'upstream' fact would require a corresponding change in the downstream fact in order to maintain consistency. We treat *causal sufficiency* (Definition 30b) as a property of situations: more precisely, as a relation which holds between a consistent situation and a fact which it contains. A situation s is causally sufficient for an effect $\langle Y, y\rangle$ if each of the (other) facts in s is necessary for $\langle Y, y\rangle$ with respect to s and these facts collectively predict the value that s actually assigns to Y when taken together with the relevant model.

(30) a. **Causal necessity** (with respect to a situation). Let $M_D = \langle D, F_D\rangle$ be a causal model over a set V of propositional variables, and let s be a causally consistent situation. Fact $\langle X, x\rangle \in \mathsf{Ker}(s)$ is **causally necessary** in s for fact $\langle Y, y\rangle \in \mathsf{Ker}(s)$ iff $X \in \mathsf{Anc}(Y)$ and for any causally consistent situation s' such that $\mathsf{Dom}(s) = \mathsf{Dom}(s')$, and $s(X) \neq s'(X)$, we have $s - s' \supseteq \{\langle X, x\rangle, \langle Y, y\rangle\}$. We write $\langle X, x\rangle \lhd_s \langle Y, y\rangle$.

b. **Causal sufficiency.** Let $M_D = \langle D, F_D\rangle$ be a causal model over a set V of propositional variables, and let s be a causally consistent situation. Situation s is **causally sufficient** for $\langle Y, y\rangle \in \mathsf{Ker}(s)$ iff for all $X \neq Y \in \mathsf{Dom}(s)$, $\langle X, s(X)\rangle \lhd_s \langle Y, y\rangle$. The set $S = \mathsf{Ker}(s)$ is a **sufficient set** for $\langle Y, y\rangle$: we write $\mathsf{Suff}_{M_D}(S, \langle Y, y\rangle)$.

Definition 30b embeds a certain notion of *minimality* (cf. Baglini and Bar-Asher Siegal 2025), in that it excludes from the set of sufficient situations for $\langle Y, y\rangle$ any situation whose domain includes variables of which Y is causally independent (i.e., any $X \in V$ such that $X \notin \mathsf{Anc}(Y)$); this follows from the necessity constraint in 30b. While it is certainly possible for a 'larger' situation s' (that is, a situation s' such that $\mathsf{Ker}(s') \nsubseteq \mathsf{Anc}(Y)$) to be causally relevant for $\langle Y, y\rangle$ in a given discourse context, this relevance can only hold where s' is a supersituation of a causally sufficient situation as defined above; Definition 30b can therefore be adopted with no meaningful loss of generality.

As defined by a model M, a sufficient situation s for fact E can be construed as a potential 'recipe' or procedure for the (actual) realization of E. It need not, however, be the *only* such procedure specified by M. The model in Figure 1, for instance, provides two (disjoint) sufficient sets for the light to be on: $\{\langle S_1, 0\rangle, \langle S_2, 0\rangle, \langle L, 1\rangle\}$ and $\{\langle S_1, 1\rangle, \langle S_2, 1\rangle, \langle L, 1\rangle\}$ (lines 5 and 9 of Figure 1b, respectively). Practically speaking, the light is expected to be on as long as one of these sets is actualized, making each set individually sufficient but unnecessary for the designated effect. The model also defines two sufficient sets for the light to be off: $\{\langle S_1, 0\rangle, \langle S_2, 1\rangle, \langle L, 0\rangle\}$ and $\{\langle S_1, 1\rangle, \langle S_2, 0\rangle, \langle L, 0\rangle\}$ (lines 6 and 8 of Figure 1b, respectively).

Within a sufficient set for effect E, necessary conditions are defined by the role they play with respect to E; as such, they can be understood as sites of potential intervention in an attempt to bring about (or prevent) E's realization. The scenario in (25b) describes one of two sufficient sets for the Lifschitz light to be off ($\langle L, 0\rangle$); the determination $\langle S_1, 0\rangle$ for S_1 is necessary for this effect within the actualized sufficient set, since any change to the value assigned to S_1 would require a change to the value of L in order to maintain consistency. The fact $\langle S_1, 0\rangle$ is thus necessary for the light to be off within a sufficient set of conditions for this result. This makes it a potential cause for the actualized situation in (25b), but (as per the discussion in Section 3.1) the truth of a claim like "Flipping switch 1 off turned the light off" would also require the actualization of both a switch-flipping and light-changing event (with the latter occurring no earlier than the former).

Taken together, definitions (30a)-(30b) echo Mackie's (1965) treatment of causality, albeit in a non-alethic setting (see Baglini and Bar-Asher Siegal 2025 for discussion). Mackie suggests that singular (single-event) *causes* can be reconstructed as so-called 'INUS' conditions, comprising Insufficient but Necessary elements of an Unnecessary but Sufficient set for their results: Definitions (30a)-(30b) give content to the INUS concept within the SEM framework. Defining necessity and sufficiency in model-theoretic terms grounds these relations in information that can be extracted from a causal model, without reference to a particular context of evaluation: they express **type-level causal** relationships, corresponding on the one hand to knowledge of how certain valuations for dependent variables in the graph can be realized, and on the other to potential truth-makers for singular causal claims in any actual context involving variables of the type included in the model.

3.3 Connecting causal models to telic predicates

On our implementation of the modeling perspective, both sufficient situations/sets and their component elements (necessary conditions) can serve as valid references for linguistic descriptions of actual causation: an episodic (singular) causal claim ϕ which links a causing eventuality C to a result E requires not only the realizations C and E, but also presupposes the existence (and cognitive availability) of a model in which C is part of a sufficient set for E. Commitment to the truth of ϕ further commits a speaker to the claim that E was realized *via* the actualization of a sufficient set containing C (with the result that ϕ is infelicitous in discourse contexts which are incompatible with the realization of the full sufficient set of which C is a necessary element). Distinct causal (causative) expressions presumably place distinct additional constraints on the role that the causing event C plays within an 'active' sufficient set for E (so that so-called 'direct' causatives might require C to complete a sufficient set, where 'indirect' causatives do not; see, e.g., Martin 2018; Nadathur and Lauer 2020; Bar-Asher Siegal et al. 2021; Baglini and Bar-Asher Siegal 2025), but—on this approach—they share crucial reference to a model in which C and E are linked by the relation in Definition 30. Singular causal claims thus rely on causal models—representations of type-causal information—for their truth and felicity, as proposed in Section 3.1.

We argue that telic predicates introduce the same sort of type-causal information as that which licenses singular causal statements. One piece of evidence for this claim was sketched in Section 2.4: where an accomplishment predicate P is used in a culminating context (e.g., in the English simple past), it not only commits the speaker to the realization of culmination condition C_P, but also to the occurrence of a process which led to C_P—that is, to the realization of some set of conditions which were collectively sufficient to realize C_P. We learn from (31) not only of the truth of the culmination condition (31b), but also of the existence and actualization of a (prior) progressive instantiation of the underlying predicate (31a), and of a causal dependence (31c) between this instantiation and the relevant culmination (cf. Szabó 2004); compare (31) to (24) .

(31) Emanuel baked a cake.

a. Emanuel engaged in some activity/series thereof. (c) $(\mathsf{PROG}(P))$

b. A cake came into being. (e)

c. c is (part of) what brought about e.

On our analysis, verifying a relation like (31c) requires reference to a causal model. This means that culminated uses of telic predicates rely on type-causal information about the conditions under which C_P is realized. In the case of singular accomplishment claims, as opposed to causative claims such as (24), verification of an actual causing event—here, the progressive-satisfying event c—also requires reference to type-level causal information: c must satisfy (part of) a fully-realized sufficient set for C_P, and the verification conditions for PROG must, as argued in Section 2.4, target procedural information about how events of type C_P are realized. Our claim is just that this procedural, type-causal information is introduced directly by an accomplishment predicate: used felicitously, any predicate in this aspectual class makes reference to a language user's normative world knowlege about how events of the culmination type are (typically) brought about.

We propose, then, that an accomplishment *event type* corresponds to a causal model, encompassing a set of generalizations about complex causal interconnections between certain conditions (point events, properties, facts, etc) which collectively comprise a set of realistic 'recipes' for C_P. An uninflected telic predicate P introduces a causal model M_P in which P's culmination condition, C_P, occurs as a dependent variable. The structure of a causal model offers a clear definition for a **culmination procedure:** the culmination procedures associated with P are simply the causal pathways—i.e., the sufficient sets—for C_P, as established by M_P. Depending on the base predicate P, the relevant model may specify a single culmination procedure S ($\mathsf{Suff}_{M_P}(S, \langle C_P, 1\rangle)$) or a range of distinct procedures (compare *run a mile* to *bake a cake*); verification of a culminated instance of P (as in 31) simply requires that one such set was realized in full (thereby verifying 31a-31c). One of the consequences of this approach—as opposed to the decompositional framework discussed in the introduction—is that the denotation of accomplishment predicates must include idiosyncratic information about the process as well as the result component of a complex telic predicate.[25]

As Section 2 explored in detail, the challenge in establishing truth conditions for progressive instantiations of telic predicates is that these conditions seem to rely simultaneously on intensional (culmination-relevant) information and a partitive concept which is concerned primarily with actual facts. The causal modeling perspective lets us cut the Gordian knot: the type-level causal information that constitutes a model is inherently intensional in nature, and reference to culmination is supplied by the notion of a culmination procedure in M_P. At the same time, a culmination procedure corresponds to a set of facts (the kernel of a sufficient situation), which means that we can compare actual, reference-time facts to such a procedure in set-theoretic terms. If the reference time facts realize part of a culmination procedure for C_P (without realizing the full procedure), they correspond to a *partial realization* of P.

The truth of a telic progressive in fact requires slightly more than partial realization—it cannot be the case that C_P has been falsified or causally precluded by reference-time facts—but this requirement, too, can be stated in model-theoretic terms: as noted in Section 3.2, the structural equations for C_P and its causal ancestors in M_P not only define sufficient situations for $\langle C_P, 1\rangle$ (culmination procedures), but also sufficient sets for $\langle C_P, 0\rangle$ (non-culmination procedures). The result, as we will spell out in the next section, is that while verification of a telic progressive requires reference to intensional objects defined by a causal model, these objects can be compared *as sets* to reference time facts, making good on the partitive intuition.

Before formalizing truth conditions for telic progressives, we reiterate the role that type-causal information plays in our analysis:

1. A telic eventuality (an event of type P) is *in progress* if some culmination procedure for C_P

[25] In standard use, the term *P-eventuality* refers to an event in the denotation of P. We adopt the term here but adjust its interpretation to the framework we are proposing: since the denotation of a telic predicate P corresponds to a causal model for C_P, we use *P-eventuality* (somewhat loosely) to refer to complete or partial realizations of culmination pathways for C_P. These realizations will often be complex, involving a variety of interacting conditions, and may well have substages or subeventualities (which in some but not all cases also validate P); in this sense, the causal modeling approach induces a causal mereological structure within the predicate itself, which then serves as the input to aspectual operators.

is partially realized (and no non-culmination procedure is complete) at reference time: as far as model M_P is concerned, then, an in-progress P-eventuality necessarily realizes a *cause* for C_P (some condition or set thereof belonging to a sufficient set for C_P).

2. The actualization of an in-progress P-eventuality neither predicts nor requires the realization of C_P itself, precisely because its causal relation to C_P is defined at the type level (in model M_P), and is not constituted by an actual causal relationship to C_P.

This is how the causal modeling perspective solves the imperfective 'paradox': since type-causal relations are generalizations, it is possible for actual events to instantiate a (normative) process for C_P cause without actually bringing this result about, and this is what permits the truthful and felicitous use of telic progressives in non-culminating contexts.

4 The imperfective paradox: the view from causal models

Truth and felicity conditions for telic progressives are given by using the tools from Section 3 to formalize the descriptive generalization (22, restated below) from Section 2.4:

(22) **Descriptive generalization for telic progressives.** Let P be an accomplishment predicate with culmination condition C_P. Then:

a. $\mathsf{PROG}(P)$ is felicitous just in case P is associated with at least one culmination procedure.

b. Where felicitous, $\mathsf{PROG}(P)$ is true just in case the situation s at reference time (a) corresponds to a (strictly) partial realization of a culmination procedure and (b) does not settle that C_P is false.

Section 3.3 defines a culmination procedure for P as a sufficient situation for the truth of C_P within a model (M_P) in which C_P is a dependent variable. Requiring P to be associated with at least one culmination procedure (as in 22a) thus amounts to presupposing that its denotation is non-empty—i.e., that there is a causal model for events of type P. Given such a model, (22b) establishes that a P-event is *in progress* at time t if the following are true: (i) some culmination procedure has been initiated at t, but (ii) is not complete, and (iii) no non-culmination procedure (i.e., a sufficient set for the falsity of C_P) is complete at t. Condition (i) ensures that (part of) a P event is realized (cf. Parsons's 1989 HOLD), while (ii) and (iii) establish that P is *ongoing* at t—specifically, that the initiated P-event has neither culminated nor terminated prior to culmination. These conditions are formalized in (32): here, s is a situation and Q, Q' are *facts* (see 28a), with $\mathsf{Var}(Q)$ selecting the underlying propositional variable and $\mathsf{Val}(Q)$ its corresponding value.

(32) **Analytical truth conditions for telic progressives.** Let M_P be a model for telic predicate P with culmination condition C_P. Given a reference time t and evaluation world w, $\mathsf{PROG}(P)$ is true at w, t just in case:

$$\begin{aligned}
&\exists s \subseteq w[\tau(s) \circ t \\
&\quad \wedge [\exists S \exists Q : \mathsf{Suff}_{M_P}(S, \langle C_P, 1\rangle) \wedge Q \in S \wedge s(\mathsf{Var}(Q)) = \mathsf{Val}(Q)] && \text{(i)} \\
&\quad \wedge [\forall S : \mathsf{Suff}_{M_P}(S, \langle C_P, 1\rangle)[\exists Q \in S : s(\mathsf{Var}(Q)) = \mathsf{Val}(Q) \rightarrow \exists Q' \in S : s(\mathsf{Var}(Q')) \neq \mathsf{Val}(Q')]] && \text{(ii)} \\
&\quad \wedge [\forall S : \mathsf{Suff}_{M_P}(S, \langle C_P, 0\rangle)[\exists Q \in S : s(\mathsf{Var}(Q)) = \mathsf{Val}(Q)]]] && \text{(iii)}
\end{aligned}$$

Although (32) expresses common-sense intuitions about what it means for an event to be in progress, it is rather complicated as stated. Moreover, it does not express clearly which elements come from the progressive operator and which from the underlying eventuality predicate. A clearer compositional picture emerges when we recognize that, as a consequence of identifying accomplishment predicates with causal models, elements that reference model structure must be provided by the predicate. Thus, while it is the progressive operator that forces instantiation of an ongoing (initiated but non-culminated and non-terminated) P-eventuality, it is the model M_P which supplies the conditions of initiation, culmination, and termination, in terms of the part-whole structure of sufficient sets for the truth and falsity of culmination condition C_P. Our proposal thus lends itself to an essentially partitive theory of grammatical aspects, wherein the progressive relates an eventuality predicate to a reference time situation in terms of *initiation* and *non-completion*:

(33) $[\![\mathsf{PROG}]\!] := \lambda w \lambda t \lambda P. \exists s \supseteq w[\tau(s) \circ t \wedge \mathsf{INIT}(s, [\![P]\!]) \wedge \neg\mathsf{END}(s, [\![P]\!])]$

Both INIT and END are defined as relations between an actual situation and a causal model M_P for predicate P. INIT checks for partial realization of a culmination procedure (corresponding to line 2 of 32), while END is satisfied if s realizes a sufficient set for either culmination (line 3 of 32) or non-culmination (line 4 of 32), so that its negation precludes any type of P-completion.

(34) Let s be a situation and P an eventuality predicate.
   a. $\mathsf{INIT}(s, [\![P]\!]) := \exists S \exists Q : \mathsf{Suff}_{M_P}(S, \langle C_P, 1\rangle) \wedge Q \in S \wedge s(\mathsf{Var}(Q)) = \mathsf{Val}(Q)$
   b. $\mathsf{END}(s, [\![P]\!]) := \mathsf{CUL}(s, [\![P]\!]) \vee \mathsf{TERM}(s, [\![P]\!])$, where
      i. $\mathsf{CUL}(s, [\![P]\!]) := \exists S : \mathsf{Suff}_{M_P}(S, \langle C_P, 1\rangle) \wedge \forall Q \in S, s(\mathsf{Var}(Q)) = \mathsf{Val}(Q)$
      ii. $\mathsf{TERM}(s, [\![P]\!]) := \exists S : \mathsf{Suff}_{M_P}(S, \langle C_P, 0\rangle) \wedge \forall Q \in S, s(\mathsf{Var}(Q)) = \mathsf{Val}(Q)$

Any aspectual operator must compose with an underlying (uninflected) eventuality predicate corresponding to a causal model. On this view, the progressive does not assert the occurrence of a process or state (Mourelatos 1978; Moens and Steedman 1988; Stout 1997), but instead that at a given time some causal pathway for P is in progress. As per (33)-(34), *being in progress* is defined by the fact that at a certain moment in time, some but crucially not all of the necessary conditions within a specific sufficient set for the culmination condition C_P are realized; the requirement against termination further ensures that, at the reference time, further progress towards C_P is possible.

Unlike on the standard approach (see Section 2.1), then, contrasts between grammatical aspects cannot be captured by merely altering the temporal relationship between some actual eventuality and the reference time. Instead, grammatical aspects contrast in *how much* of a causal pathway in the event predicate model can or must be realized by a reference-time situation (i.e., by differences with respect to the value of CUL and/or TERM). While a full-fledged partitive theory of aspect is beyond the scope of the current paper, we believe that our treatment of accomplishments offers a straightforward account of the *non-culminating perfective* uses which have been observed in a wide range of languages (see Martin, 2019, and references therein).[26] The relations in (34) should also play a role in the decompositional analysis of aspectual verbs such as *begin, continue,* and *stop*, discussed in the introduction. We leave this as a topic for future investigation.

[26]The idea is that culminating and non-culminating perfectives are unified against progressives by a shared END requirement, but differ (within and across languages) in whether they are specified for END (34b) or CUL (34b-i). A 'strong', culminating perfective (such as the English simple past or the Hindi compound perfective; see Arunachalam and Kothari 2011; Singh 1998) would require a situation fully contained within reference time to verify both INIT (34a) and CUL (34b-i), whereas a 'weak', non-culminating perfective (such as the Hindi simple perfective) specifies INIT and END (34b), and is thus satisfied in contexts where a telic eventuality is initiated but ends either by culminating or by terminating prior to successful culmination (see Altshuler 2014; Nadathur and Filip 2021).

4.1 Revisiting the data

Our discussion of intensional Type I approaches to the imperfective paradox focused on three types of data: progressives of impossible tasks (IEs), unexpected events (UEs) and telic progressives in out of reach contexts (ORCs). While normality-based approaches (e.g., Dowty 1979; Asher 1992) predict automatic falsity for all three types of progressive on the grounds that the relevant reference contexts in each case fail to predict culmination, we fine that IE progressives are empirically distinguished from UE and ORC progressives (see Table 1 below). IE progressives are uniformly rejected, but UE and ORC progressives can be either true or false, depending on what is actually going on at reference time. Our intuition, which we used to motivate the causal modeling approach, is that the key difference has to do with the nature of the underlying predicate: IE progressives involve tasks which are categorically unrealizable, while UE and ORC progressives do not.

Data type	Examples	Empirically	Type I prediction
IEs	(16a) Mary is swimming across the Atlantic Ocean (16b) The children are digging a hole to China	X	false
UEs	(17) Trapper John was crossing a minefield (18) 100 boats are circumnavigating the globe	✓	false
ORCs	(19) Benny was running an ultramarathon (1) The architect was building the cathedral	✓	false

Table 1: Intensional progressive (Type I) predictions for IE, UE, and ORC progressives

By relying on an association between telic predicates and type-level causal models, the truth conditions in (32)/(33) make good on this intuition, and divide IEs from UEs and ORCs as desired. On our analysis, IE progressives like (16a)-(16b) are rejected due to infelicity, rather than strict falsity. Use of a telic predicate P requires the existence of corresponding causal model, establishing at least one cognitively-plausible procedure for realizing culmination condition C_P: a predicate which describes a task which is known to be impossible (such as swimming across the Atlantic Ocean or digging a hole through the earth to China) is, *ipso facto*, a predicate for which the type of world knowledge which comes into play in the construction of an event type causal model litigates against the existence of such a pathway. In the absence of a model, there is no object against which reference time facts can be evaluated in the terms set out in (33), and IE progressives are therefore not truth-conditionally evaluable.

To the extent that the details of a causal model are supplied by world knowledge, the epistemic state of a language user plays an important role in the evaluation and evaluability of a telic progressive. Use of a particular predicate presupposes the existence of an appropriately-specified causal model in the following sense: it signals the speaker's belief in the existence of the model, and their belief that such a model is shared (to some level of granularity) with their interlocutors.[27] IE progressives are rejected due to the absence of a shared model: the audience for, e.g., (16b) is unlikely to accept the speaker's apparent presupposition (that there is a way for the children to dig

[27] We suggest that the default assumption, in encountering a novel telic predicate, is to take the user's state as authoritative: that is, in the absence of world knowledge which cuts against the existence of an appropriate model (as in the IE cases), a hearer is likely to accept the use of an unfamiliar predicate as felicitous.

a hole to China) precisely because it is recognizably absurd for a well-informed adult to believe in the achievability of the underlying task. Things can change if allowances are made for the speaker's epistemic state, as the scenario in (35) shows:

(35) **The hole-digging scenario revisited.** Meena's five year old daughter Maya wrongly believes that the earth is made entirely of sand and soil, and that it has a considerably smaller diameter than in actuality. Maya is digging a hole at the beach, with the stated intention of tunneling all the way through the planet.

a. *Maya:* I am digging a hole to China!

b. *Meena:* ??Maya is digging a hole to China.

In this context, the child's epistemic state—which crucially contains false beliefs about the nature of physical reality—supports the existence of a fairly straightforward strategy for digging a hole to China, and thus the existence of a plausible type-level causal model for this task. Although Maya is objectively misguided—and while it would clearly be inappropriate for the adult observer to sincerely describe Maya's actions with (35b)—it does not seem appropriate to say that she is lying: (35a) is both an appropriate and true use of the progressive, as far as Maya is concerned. The causal modeling approach allows us to make sense of the contrast: an objective and well-informed (adult) observer will not treat Maya's description of her own behaviour as authoritative or grounded in reality, but also will not respond to it as inappropriate, since the conversation always relies on the individual speaker's world knowledge.[28]

Unlike IEs, UEs and ORCs both involve predicates which are generally accepted to be possible (if perhaps non-trivial) to fully realize, and which thus have event type models as far as the typical language user is concerned. Progressives of these predicates can simply be true or false of a given situation, depending on the validity of all three conditions in (32)/(33); there is no meaningful difference between UEs and ORCs on the causal approach. In the example of the undertrained runner, for instance, (19a) will be true as long as Benny has started the race (e.g., by taking a step into the race path after the starting gun has fired; INIT), has not reached the end of the race path ($\neg\mathsf{CUL}$), and is able at reference time to continue towards completion ($\neg\mathsf{TERM}$). Crucially, the truth of the progressive does not take into account the likelihood that Benny will successfully complete the race: once he has initiated an appropriate process, all that matters at a given moment is that it remains possible for him to make further progress. This possibility collapses as soon as Benny's stamina runs out, but not before: in the terms in (32)/(33), the total depletion of energy which accompanies his inevitable collapse completes a sufficient set of conditions for non-culmination, validating TERM (34b-ii) and thus falsifying the $\neg\mathsf{END}$ requirement of (33).

[28] The relationship between causal models and world knowledge also predicts that it is possible for two speakers to agree on the existence of a causal model for a predicate P but to disagree on its granular details. The result is that speakers may disagree on the truth of a given progressive without either deviating from a valid evaluation of the conditions in Proposal (33), relative to their own information state. Suppose, by way of illustration, that Meena is making an angel food cake (which contains no butter), and suppose further that she is at the point of putting her batter into the oven. Observer A, whose cake-baking model allows for angel food cakes, could truthfully claim that *Meena is baking a cake*, but an observer B who believes that all cakes require butter could not. There is of course an objective truth of the matter, according to which B is wrong, but once we understand how his model is constructed, we do not want to say that B is being disingenuous in rejecting the target progressive. The causal modeling approach allows us to explain this 'faultless disagreement' along the same lines as the perspective shift in (35).

4.2 A revision: globally necessary conditions

It is, of course, possible for Benny's bid to run an ultramarathon to terminate unsuccessfully even if his energy is never fully depleted: this is just one way that a pathway for non-culmination may be completed. Other events which close off the possibility of further progress—such as a broken or sprained limb, or official termination of the race due to inclement weather—can likewise complete a sufficient set of conditions for non-culmination. One possibility holds a special status with respect to a plausible causal model for ultramarathon running. Assuming that running any official race is an agentive (volitional) pursuit, an agent's intention to realize the relevant culmination condition C_P acts as a **globally necessary condition** for culmination: that is, as a condition which must not only hold throughout the development of any causal pathway for culmination, but which moreover sustains and guides this development (cf. Varasdi 2014 on *sustaining conditions*). Formally, a globally necessary condition can be modeled as a proposition whose falsity acts as a singleton sufficient set for non-culmination, as spelled out in (36):

(36) **Globally necessary conditions.** Fact Q is *globally necessary* for C_P with respect to model M_P just in case $\mathsf{Suff}_{M_P}(\{\neg Q\}, \langle C_P, 0\rangle)$.[29]

Based on the non-termination requirement of (33), we predict that the truth of an in-progress accomplishment can be toggled on and off with changes in an agent's culmination-relevant intentions: the judgments we have collected for the following running scenario support this prediction.

(37) **The (re)commitment scenario.** Benny began running in a marathon at 9:00am on an extremely hot day. At 11:35am he felt overheated and exhausted, and stopped at an aid tent with the intention of quitting the race then and there. After a short rest and some refreshment in the shade, however, he started to feel better, and so at 11:48 he decided to finish his race. He started running again at 11:50am.

a. *At 11:30, 11:55:* Benny is running a marathon.

b. *At 11:45:* #Benny is running a marathon.

As long as Benny's intention to run a marathon is modeled as a GNC, its falsity is enough to ensure the falsity of the present progressive at the time specified in (37b) (as well as at any other point between 11:35 and 11:48am). What is particularly interesting about this is that the truth of the progressive reverts once Benny's intention is reactivated: his recommitment to the race at 11:48 not only makes (37a) true at 11:50, but moreover recasts Benny's activities between 11:35 and 11:48 as part of a durative eventuality which verifies the predicate $\sqrt{}$*Benny run a marathon*.[30]

It turns out that GNCs also have a deeper role to play in the evaluation of telic progressives: their determination appears to form a minimal requirement for the truth-conditional assessment of a telic progressive, even if certain other conditions which are plausibly part of a causal pathway for

[29] We use $\neg Q$ to represent the ordered variable-value pair which shares the variable coordinate of a fact Q but flips its value coordinate from 1 or 0 (or vice versa). Since the fact Q corresponds to a standard proposition, this notation reflects standard logical negation.

[30] It is worth noting that manner characterizations, at least for telic predicates which specify a manner of motion for their process components, are not necessarily identified as GNCs. It is possible, for instance, to be running a marathon while one is not running at all—for instance, while one is drinking water at an aid station, or even taking a brief sitting rest by the side of the race course. Pauses of this sort need not affect the satisfaction of the truth conditions in (32)/(33), since they need not falsify INIT or validate END (and they do not preclude the possibility of further development along an appropriately specified causal pathway for the culmination condition.

culmination have already been realized. This effect can be seen, again in the context of agentive accomplishments, by considering yet another variation on the running context:

(38) **The variable distance scenario.** Benny began running in a race which had different distance options: 10 miles, half marathon, and full marathon, with separate finish lines along the main course. Uncertain about his precise level of preparedness, Benny did not intend to complete the full marathon distance, but planned to decide at the 10 mile turnoff whether to stop there or continue to run a half marathon. He unexpectedly collapsed between the 8th and 9th mile marker, before making a decision. Later, he says:

a. ??I was running 10 miles when I collapsed.

b. ??I was running a half marathon when I collapsed.

c. ✓ I was running 10 miles or a half marathon when I collapsed.

d. X I was running a marathon when I collapsed.

In context, Benny's intentions explicitly preclude running the full marathon, rendering (38d) false despite the fact that he has taken steps along the race path. Since he has the disjunctive intention to run 10 miles or a half marathon, (38c) is true: up to the moment of collapse, he was on a pathway for culmination (of one of these tasks) which was neither completed nor terminated. However, since Benny lacks both the specific intention to run exactly 10 miles and the intention to run a half marathon, neither (38a) nor (38b) seems to be an appropriate description of his activities. We can capture these judgments by constraining our analysis of telic progressives as in (39).[31]

(39) **Determination of GNCs.** $\mathsf{PROG}(P)$ is defined at $\langle w, t\rangle$ iff no GNCs for C_P are undetermined in context: $[\![\mathsf{PROG}(P)]\!]^{w,t} \in \{0,1\} \leftrightarrow \forall Q : \mathsf{Suff}_{M_P}(\{\neg Q\}, \langle C_P, 0\rangle), s(\mathsf{Var}(Q)) \neq u$

We focus here on agentive accomplishments, since intent for culmination is a clear example of a GNC in these cases. However, non-agentive predicates also have GNCs, which should—if we are on the right track—play the same role as intentions with respect to the truth-conditional evaluation of telic progressives. Appropriate conditions in the non-agentive cases seem to involve properties which sustain the development of a P-eventuality from initiation through completion (cf. Varasdi 2014): that is, which 'guide' it towards culmination in a manner analogous to agentive intent. Reasonable candidates, then, might be properties like momentum or other conserved quantities (consider, for instance, a non-agentive accomplishment such as $\sqrt{}$*The ball roll off the table*). Knowledge of such quantities—unlike knowledge of intentions, which must be inferred from agents' behaviour or accepted purely based on self-report—is a matter of fine-grained physical measurement and observation, to a degree which may not be possible for the casual observer. This leads to an interesting predicted contrast between agentive and non-agentive progressives. On our proposal, Benny's underdetermined intention in (38) is sacrosanct: there is simply no truth of the matter with respect to (38a) or (38b) until and unless he makes up his mind. In a parallel non-agentive case, such as (40), the progressive of one of the disjuncts is ostensibly true, but its determination is out of reach of the standard (unaided) human observer.

(40) A fair coin is tossed, and is still rising.

a. ?The coin is coming up heads.

[31] This is a conservative form of the GNC constraint. We do not think that (39) is peculiar to telic progressives, but instead represents a restriction on the predicate-internal INIT condition. We refrain from taking a strong position here, as we have yet to investigate the relationship between GNCs and non-progressive uses of telic predicates.

b. ?The coin is coming up tails.

(41) a. *In (38):* Benny is running a 10 mile race or a half marathon, ??but I don't know which.

b. *In (40):* The coin is coming up heads or tails, but I don't know which.

This prediction—and thus the causal approach to telic progressives—gains some preliminary support from the contrasting judgments in (41), but a detailed investigation is left for future research.

Finally, an interesting and positive consequence of (39) is a straightforward account of a surprising temporal asymmetry in the truth and felicity of telic progressives, first noted by Bonomi (1997): past tense progressives can sometimes be verified with respect to a reference time at which the corresponding present tense claim could not have been truthfully uttered. For instance, if we modify the scenario in (38) so that Benny reaches the 10 mile mark and decides to stop there, the past tense claim in (42a) becomes true for any reference time between the time at which he started the race and the time at which he stopped running; (42b) would not have been true if uttered at any of those times. Unlike most Type I approaches to telic progressives, the GNC presupposition in (39) predicts this asymmetry, since a relevant GNC (here, Benny's distance-relevant intention) was not determined until the 10 mile mark. GNCs, in other words, represent necessary conditions for the conceptual initialization of a process for C_P, but their determination need not correspond to the temporal initiation of such a process.

(42) a. Benny was running 10 miles.

b. Benny is running 10 miles.

The asymmetry can also emerge in another way: we find that the empirical judgments of (42a)-(42b) are the same if Benny reaches the 10 mile marker and collapses without making a concrete decision about his goals. This suggests, at least at first glance, that agents' intentions are relevant to judgments of telic progressives only from epistemic perspectives at which the outcome of the event in progress remains undetermined—e.g., in the present tense, but not in the past.

The causal modeling perspective also offers an explanation of this contrast, as long as we accept that certain accomplishments—e.g., running a particular distance—can be realized both as deliberate, *agentive* acts (wherein goal-directed intentions act as GNCs) or as non-agentive actions, and that the truth conditions for these realizations may well differ (see, e.g., Belnap and Perloff 1988). When Benny completes a 10 mile run and then stops unintentionally, the model for the agentive form of the underlying predicate becomes immediately irrelevant, as his prior actions can be identified directly as belonging to a realization of a (non-intentional) culmination pathway for a 10 mile run; the relevant *télos* (and thus the relevant model) is picked out by actual events.[32] While the process is still unfolding, however, truth-conditional judgments of the progressive rely on being able to identify the appropriate model for comparison: scenarios which allow for multiple outcomes (such as the running scenarios under discussion here; see also Bonomi 1997 on the *problem of underdetermination*), only evidence of the agent's intent (goal) can single out a specific model from among relevant alternatives (cf. Varasdi 2014 on *indicative conditions*). If this intention is absent—whether it is underdetermined or simply inaccessible to the speaker—truth conditional assessments of telic progressives are impossible until enough evidence has been gathered to supply a predicate model for comparison.

[32]This type of explanation is indeed foreshadowed by Vendler (1957, p.146) in his discussion of goal-oriented predicates: "[Accomplishments] proceed toward a terminus which is logically necessary to their being what they are. Somehow this climax casts its shadow backward, giving a new color to all that went before."

This also explains why exactly one of the past tense correlates of (40a) and (40b) becomes true (and the other false) as soon as the coin has landed. The difficulty in asserting the present tense claims arises from the epistemic inaccessibility of fine-grained measurements which (in a physical sense) determine the outcome while the event is in progress; this information can only be gleaned retrospectively and indirectly once the coin's trajectory has culminated.

4.3 Causal models and event structure

Type I and P approaches to the imperfective paradox (see Section 2) are differentiated on the basis of intensionality: Type I treats 'paradox' effects as the consequence of modality introduced by the progressive operator, while partitive Type P approaches pursue an extensional account of the data. Insofar as our account involves direct evaluation of the part-whole relationship between actual events and (causal pathways in) the denotation of a telic predicate, we take a broadly partitive view of the puzzle: our account is not, however, an extensional one. While events that fall short of culmination can validly instantiate a telic event type, they must do so by virtue of correspondence to a (culiminating) whole defined by a type-level causal model: to the extent that such a model expresses generalizations about how events of a given type are ordinarily realized, it is an inherently intensional object. Our account thus shifts the modality associated with telic non-culmination from the contribution of an aspectual operator to the denotations of telic predicates (cf. Koenig and Muansuwan 2000; Nadathur and Filip 2021).

Causal models capture information about what is causally normal; our proposal thus shares with Type I approaches a reliance on intuitions about stereotypicality. Shifting the anchor of these intuitions from the progressive to the underlying predicate has a substantive effect on their role. Where Type I approaches ground normality in an evaluation context, the causal modeling perspective ties normality to the lexically-specified culmination condition of a predicate: for us, the crucial consideration is not whether the reference situation locally predicts C_P, but whether it corresponds to a plausible cross-section of some normative pathway for C_P. The causal modeling approach asks the following question: does the reference time situation correspond to an expected *developmental stage* of a maximal P-eventuality—that is, does it match *what could be happening in an unremarkable world* where C_P gets realized?[33] This perspective allows us to explain UE and ORC progressives, where traditional Type I analyses do not: a given state of affairs can correspond to a normal developmental stage of a hypothetical culminated P-eventuality without predicting culmination in the context in which it is actually embedded.

In taking a partitive but still intensional view, our proposal has something in common with Type I analyses that take the internal structure of complex events into account. As we will see, reference to (mereological) event structure allows Type I accounts like those of Landman (1992) and Bonomi (1997) to fare better than 'pure' Type I approaches (see Section 2.2) on UE data. However, we argue that they ultimately fall short on the treatment of ORC progressives, as a consequence of maintaining reliance on a set of modal alternatives locally projected from the context of evaluation.

Landman (1992) introduces a primitive notion of *event stages* (wherein "an [actual] event is a stage of another [actual] event if the second can be regarded as a more developed version of the first"; p.23) in order to treat the progressive as a relation between an actual event e and a

[33]This is reminiscent of teleological modality, which is concerned with what is necessary or possible in worlds where some relevant goal or *télos* is realized (see, e.g., von Fintel and Iatridou 2005): a causal model for predicate P can be conceptualized as a description of a set of ideal, causally normal worlds in which the *télos* C_P is realized. See Nadathur and Filip 2021 for further exploration of this idea.

predicate P: $\mathsf{PROG}(P, e, w)$ is true just in case e is a stage of a (culminated) P-eventuality in the **continuation branch** of e with respect to w. The crux of the account is the notion of a continuation branch, which tracks event e across a set of 'reasonable' (plausible) alternatives to w: roughly speaking, we follow e to the point at which it ceases in w, and if this is due to some external interruption to development, we shift to its continuation in the nearest reasonable alternative where the interruption does not occur. The process is repeated until the tracked event comes to a natural culmination or its continuation moves beyond the 'reasonability' horizon.

One advantage of the continuation branch approach over the Type I proposals discussed in Section 2 is that interruptions need not be treated as abnormal in any objective sense: for Landman, $\mathsf{PROG}(P, e, w)$ holds just in case there is a reasonable chance that the processes defining an actual event e will continue to develop towards C_P at each point at which they might be interrupted. UE progressives can then be verified because the continuation branch 'removes' potential interruptions sequentially (Landman, 1992, p.30)—all that is required is that each potential point of disruption (however likely it may be) might plausibly be avoided, allowing the event to continue for a little bit longer. The appeal to reasonability also rules out IE progressives like (16a), as desired: even if Mary gets into the ocean at Galway (event e), and even if there is one reasonable alternative in which she continues swimming westwards, drawing on a last reserve of energy, past the point at which she actually succumbs to exhaustion, doing so at every potential failure point will (per Landman) move us outside the set of reasonable modal alternatives well before the crossing is complete, so that the continuation branch of e cannot verify the progressive with respect to the target predicate.[34]

(16a) #/??Mary is swimming across the Atlantic Ocean.

Like our analysis, Landman's approach centers a conceptualization of actual events as a developmental stage of a full P-eventuality, and prioritizes the possibility of continuation over an expectation that culmination will take place. This allows him to make the right predictions for UE progressives; unfortunately, as Bonomi (1997) points out, it leads to the wrong predictions for 'multiple choice' scenarios like (43).[35]

[34]Landman (1992) is partly motivated by the need to balance the "problem of interruptions" with what he calls the "problem of non-interruptions", referring to a shift in judgments observed in case an impossible task is actually completed. In such a scenario—e.g, where we learn from a credible witness that Mary has just climbed out of the ocean in Newfoundland—the past tense correlate of (16a) becomes true, even though (16a) would not have been accepted at the same reference time. As discussed at the end of Section 4.2, standard Type I approaches struggle to account for this type of temporal asymmetry, but Landman sidesteps the problem. Since the continuation branch first follows an actual event e to its maximal development, the fact that Mary has actually succeeded in crossing the Atlantic cannot be ignored: the culminating continuation branch is fully contained in the evaluation world, and what actually happens is enough to verify the (past) progressive.

We have already previewed how the causal perspective handles this data. The underlying intuition is very similar to Landman's: whatever actually happens constitutes a course of events that is, *ipso facto*, causally normal. We assume that the actual world is treated as causally normal (i.e., that causal reasoning employs as a default that "anything that has happened must be possible"), so that knowledge of Mary's fluky success forces the conclusion that some sufficient set of causal conditions for culmination was actually realized, and thus that there is a plausible causal pathway: it is true that *Mary was swimming across the Atlantic* because the actual world demonstrates not only that there is a way to do so, but also that Mary was (at reference time), engaged in the right sort of process.

[35]Insofar as the continuation branch must culminate within the set of reasonable options (projected from world w and event e), Landman's PROG is ultimately an existential intensional operator. We showed in Section 2.2 that such an analysis could not explain the acceptability of ORC progressives: indeed, it is difficult to see how Landman's explanation for the failure of IE claims—that culmination simply crosses the reasonability horizon—would not also apply to the original scenario (19) of the undertrained runner. Reliance on the existence of an underlying causal model (to provide the truth conditions for a valid event stage) allows us to make the necessary distinction between

(43) **The multiple-choice scenario.** Meena is driving from Monterey towards San Jose. Her intention is to go to either San Francisco or Oakland, but she has not yet decided which.

a. Meena is driving to San Francisco or Oakland.

b. ??Meena is driving to San Francisco.

c. ??Meena is driving to Oakland.

In this scenario (as in 38), the disjunctive progressive is intuitively true, but neither of the disjuncts is appropriate. The causal modeling approach explains these judgments with reference to GNCs, as dicussed in Section 4.2: (43) satisfies the intention GNC for (43a) but not for (43b) or (43c); moreover, there does not appear to be an underlying truth of the matter. On Landman's analysis, however, the truth of (43a) requires that one of (43b) or (43c) is true (even if we do not know which): the culmination branch must verify a completed trip to San Francisco or Oakland—that is, Meena's journey must end in exactly one of these two distinct locations.

Bonomi's own analysis avoids the problem of disjunctive outcomes by adopting a context-sensitive modal analysis which incorporates a mereological structure for event types (p.188). The underlying intuition is that an actual event can be seen as a part (or stage) of a range of distinct, more developed eventualities, but that which—if any—of these larger events is realized depends on the context in which the reference event occurs. Bonomi proposes that the progressive expresses an intensional relation between an actual event e and a predicate P by projecting a set of "courses of events" (akin to Thomason's 1984 *histories*) from e, taken together with some contextual set C_H of facts. These courses of events are stereotypical in the sense that they permit e to follow its natural development: $\mathsf{PROG}(e, P)$ is true relative to C_H just in case all of the "ideal" developments of e which conform to C_H contain an event f of type P such that e is a stage of f.

This treatment explains the judgments in (43) if we take the reference event to comprise Meena's trip-related activities, including information about her point of origin, her route thus far, her speed, direction of travel, and so on. This event could be seen as part of any number of northward journeys, but its development is constrained by local ("concomitant") facts in C_H such as the amount of gas Meena put into the car, how far she is willing to drive, and any intentions she has with respect to her destination. Including Meena's intention to drive to either San Francisco or Oakland in the constraining set C_H restricts the set of stereotypical developments of her trip thus far to include journeys which end in either of the two cities, validating (43a) but falsifying (43b) and (43c) (since the relevant contextual facts do not successfully pick out a single destination).

Per Bonomi, UE progressives can be true because the alternatives projected by PROG are relativized to the reference event e. He argues that the "ideal" courses of events exclude "all external factors that might block the completion of the event" (p.190), taking into consideration only what has actually happened thus far as well as concomitant facts that have to do with direct participants in the named event. This is similar to our treatment of UE progressives in prioritizing information about what is actually going on at reference time, but is differentiated by a selective reliance on context—in this case, on positive, supposedly event-external information which supports the idea that successful culmination is *possible* (if not especially likely).

This context-dependence makes it difficult to extend Bonomi's account to a successful treatment of ORC cases: it is not clear which contextual information should count as event-internal and event-external but relevant as a concomitant fact as opposed to a potential interruptor.[36] Returning to

IEs and ORCs, as laid out in Section 4.2, where reliance on what is reasonable on the basis of local facts does not.

[36]Bonomi (1997) motivates the introduction of an event mereology by observing that neither a requirement of

the case of Benny the runner, excluding his (under)preparedness from the relevant concomitant facts—as a fact which might block the continued development of a running event—seems to allow the truth of (19a), as desired; however, it is not at all clear that this information can be innocently excluded from C_H given the context provided in (19). Indeed, reference to concomitant facts of this sort suggest that Bonomi ultimately relies on the idea that some course of events in the stereotypical set is a possible future with respect to the local context in which e is embedded.

(19) **The undertrained runner** (modified from Varasdi 2014, pp.192–193). Benny is an amateur but ambitious runner who decides to run an ultramarathon. He will not in fact be able to run the whole distance, because very few can, and he does not have the necessary physical qualities, having trained insufficiently to build up his endurance to the required degree. Nevertheless, he registers for the race, appears on the given date, gets his race number, and sets off with the other runners. The first couple of kilometers go well, but then he begins to lose strength, and at about half the distance he collapses because of exhaustion.

a. *Target:* Benny was running an ultramarathon (when he collapsed).

To make matters worse, the processes in which Benny is involved after the race begins very reasonably include a process of energy depletion, so that even stereotypical continuations of the reference event e should reasonably prevent it from being seen as part of a larger, ultramarathon-completing eventuality. By fully divorcing notions of event structure from the local context—and by evaluating the progressive as a relation between a set of reference-time facts and an event-type model—the causal approach simplifies the calculation here: we do not need to consider the relative speed and trajectory of actual ongoing processes, but simply whether not a set of instantaneous facts (Benny's intentions, his placement along the race course, that he has at least some energy at reference time) correspond to some stage in the development of an idealized ultramarathon-completion.

The account which bears the closest resemblance to ours is that of Varasdi (2014). In view of examples like (19), Varasdi argues—as we have—that the truth of a progressive claim involving telic predicate P does not require that culmination is a possibility as assessed from the reference time perspective. Instead, he proposes to treat the progressive as a property of states, which can be true of a reference situation s just in case s verifies conditions which are necessary for a (culminated) event of type P. The analysis is motivated by the asymmetric entailment relationship between a (culminating) telic perfective and its corresponding progressive (see Section 2), which (cf. Szabó 2004) shows that the progressive of a predicate P is verified by a strict subset of the conditions that make its perfective counterpart true.

Varasdi proposes that a context in which $\mathsf{PROG}(P)$ is used implicitly makes available a set R of relevant distinct outcomes which includes (the culmination condition of) P. The progressive claim is true in such a context just in case there is a particular facet F (subset of facts) of the reference situation s) which is *indicative* of P in R—i.e., such that the characteristics of s which

temporal nor causal precedence (between the actual event e and the culmination of P) alone is enough to justify the use of the progressive: for instance, while all events which culminate in a painted house must be both temporally and causally preceded by events in which some agent acquires the necessary painting materials, an event in which this agent purchases the required equipment will not in general verify the progressive with respect to house-painting. Introducing an atomic notion of event parthood pushes this issue into a question of world knowledge: while it might not be straightforward to specify general conditions on the activities A which constitute an initiated part of a predicate P, common sense judgments supply information on a case-by-case basis. This is the type of information that is presumably encoded in a causal model, or—for Bonomi—which govern the part-whole relations between events in a stereotypical set of histories.

are picked out by F are only compatible with P. This is operationalized via three conditions in a possible-worlds framework: first, that some world w which realizes the facet F of s is such that some mereological development of s (i.e., some event of which s is a part) in w is an event of type P; secondly, that no such world realizes an alternative outcome from the set R as a mereological development of s; and finally, that worlds which do not verify F as a facet of s are worlds in which P is not realized, making the conditions in F necessary (albeit not necessary sufficient) for P.

Although Varasdi does not discuss IE progressives directly, the account should treat these as false (rather than infelicitous), provided that the set of worlds from which the perspective verifying F as a facet of s is drawn are restricted to worlds which are broadly causally normal: under this assumption, while the perspective picked out by s and F may well be non-empty, none of them can contain a full realization of the IE predicate, and thus the first condition for a telic progressive fails. By virtue of being achievable in causally normal albeit atypical circumstances, UE progressives can be satisfied by a state of affairs just in case there is some perspective which verifies a set of necessary conditions for the underlying predicate P which uniquely select from the set of contextually-relevant outcomes. IE progressives work much the same way: as above, we need to abstract away from local facts—such as Benny's ill-preparedness—which preclude the right sort of culmination, but this exclusion can occur as part of the selection of a given facet/perspective on the reference situation, and need not be justified by considerations of what is internal/external to an ongoing reference-time event, as for Bonomi's account.

The conceptual resemblance between Varasdi's proposal and ours is immediate. Both analyses prioritize the information included in a reference-time state of affairs over the contextually-expected continuations of this state, with the result that UE and ORC progressives can be true. In both cases, the idea that the evaluation context must realize *sufficiently much* of an event of type P is reduced to a type of subset relation: as Varasdi notes, necessary conditions are (can be) sufficient. Despite these similarities, the accounts differ in two consequential ways. First, Varasdi relies on the idea that necessary conditions are only enough to verify a predicate in case these conditions indicate P out of a set of contextually-relevant outcomes: as far as we can tell, this causes problems for progressives with disjunctive outcomes, as in (43). It seems reasonable from a pragmatic perspective that the individual predicates $\sqrt{}$*Meena drive to San Francisco* and $\sqrt{}$*Meena drive to Oakland* are included in the relevant set of contextual outcomes whenever $\sqrt{}$*Meena drive to San Francisco or Oakland* is; however, if all three alternatives are included, then it should not be possible to select a facet of any reference time situation which admits events verifying the disjunctive outcome without also admitting events verifying one or the other of the individual disjuncts. Secondly, while both our account and Varasdi's require that the reference situation contains some set of facts which are necessary for a full realization of predicate P, Varasdi's reliance on alethic necessity (i.e., strict entailment) implicitly requires that there is only one way—one set of sufficient conditions—which can fully realize a given predicate P. While this may be appropriate for predicates with relatively homogeneous process components (*run a marathon, climb the stairs*), it seems less appropriate for predicates whose culmination conditions may be realized in a variety of distinct ways (*bake a cake*, etc); defining necessary conditions with respect to their role in potentially distinct sufficient sets in an event type causal model, as we do in Section 3.3 allows for this possibility. It may of course be possible to amend Varasdi's account to handle both of these issues, but—to the extent that his account must independently incorporate a mereological event structure according to which a given state can be seen as part of a larger eventuality—our view is that the causal modeling approach streamlines and simplifies the implementation of the shared underlying ideas.

5 Conclusions and outlook

This paper is organized around a lexical puzzle associated with (durative) telic predicates: how the clear and intuitive comnnection between such predicates and their lexically-specified culmination conditions can be reconciled with their robust use in non-culminating contexts. Using the well-known imperfective paradox as a case study, we identified the main theoretical challenge in explicating telic non-culmination as that of establishing a satisfactory notion of *partial realization*—that is, in establishing formal conditions under which a non-culminated eventuality realizes "sufficiently much" of the telic event type to verify the use of the underlying predicate: in the context of the progressive, these are the conditions which qualify an actual eventuality as one which "makes progress" towards culmination.

On the received (Type I) approach to the paradox, intensionalizing the progressive operator allows partial realization to be cashed out indirectly, in terms of the "normal" or expected (modally-projected) consequences of a reference time situation. Building on observations from Bonomi (1997); Szabó (2004); Varasdi (2014), and others, we argued in Section 2 that these approaches cannot adequately explain empirical judgments for progressives of unexpected events and predicates whose culmination is ordinarily realizable but locally out of reach. Based on these data, we argued that so-called 'paradox' effects cannot be handled by means of an intensional progressive operator, proposing instead that intensionality enters into telic non-culmination phenomena through the need to compare actual facts to some idealized (normative, hence intensional) notion of a culmination recipe or procedure. As Section 3 lays out in detail, this notion is supplied by equating a telic predicate with an event type causal model, so that the denotation of the predicate supplies one or more sets of collectively sufficient conditions for bringing about the relevant culmination condition. Treating a telic event type P as a causal model permits straightforward evaluation of a reference time situation (set of facts) as a potential partial P-eventuality, by virtue of a truth-conditional match between the set of facts comprising the token and a cross0-section of a normative process for P, as established by model M_P (a type-level representation of how culmination condition C_P is ordinarily brought about).

While our proposal links telic predicates to inherent causal structure, we emphasize that the causal modeling approach diverges from a classical causative (CAUSE-predicating; see Section 1) analysis of the telic aspectual class (cf. Dowty 1979). As discussed in Section 3, CAUSE-decompositions necessarily link some causing event involving the sentential subject to the actual realization of the culmination condition in a (uniform) binary causal relationship, with the result that the underlying predication of actual causation can only be validated in case both process and result components of the predicate are actualized in the evaluation context. While we agree that a telic predicate does indeed invoke causal semantic structure, whereby some process (set of conditions) are linked at the type level to a lexically-specified culmination condition, our claim is that the details of this process, as well as the nature of the causal relationship between individual process steps and culmination, are supplied by world knowledge, as is an assessment of whether or not a causal model with the appropriate lexical specifications is cognitively plausible in the first place. Use of a telic progressive conveys a speaker's knowledge of (belief in) a valid causal model for the underlying uninflected predicate, but neither predicates an actual causal relationship (to culmination) not amounts to a modal prediction of culmnination by some other means; instead, it simply asserts (reports on) a perceived match between actual facts and some developmental stage in a process/recipe for culmination, as provided by the type-level model. All that is required for P to be in progress at time t is that the set of actual facts at t are compatible with some stage in a normative process for C_P, and that—at the reference time—nothing precludes further development towards C_P. In this approach, a process is conceived as a causally structured set of states and events that together constitute a sufficient pathway leading to a culminating event, rather than as a distinct entity or a mere partial stage.

Alongside the formal felicity and truth conditions for telic progressives provided in Section 4, we suggested that treating complex, result-oriented events types in terms of causal models induces a causal mereological structure which can be extended beyond the imperfective paradox to provide parallel formal accounts of telic non-culmination in the scope of other aspectual operators, such as non-culminating perfectives (see Section 4.1) or aspectual verbs (see Section 1): an approach which combines a causal modeling treatment of telic predicates with an account of aspectual operators as essentially partitive operators (see, e.g., Altshuler 2014; Nadathur and Filip 2021) in the crosslinguistic perspective is thus a fruitful avenue for further exploration. Looking farther afield, we suspect that telic predicates are not the only aspectual class whose internal structure and properties can be fruitfully treated in terms of causal models: in this sense, the success of the causal modeling approach in explicating judgments of telic progressives is simply presented as a motivating study for a more extensive causal approach to the structure of events.[37] Dowty's (1979) aspect calculus—which originally gave rise to the causative analysis of complex event types—was originally motivated by a desire to explain the patterns and contrasts between aspectual classes in terms of a small core set of structural features: our proposal is that this project may be fruitfully undertaken by means of causal models instead. This paper then lays the groundwork for a broader theory on which predicates of eventualities are labels for collections of causal conditions which enable, allow, sustain, constitute, or otherwise lead to particular outcomes or states, and the familiar aspectual class properties (durativity, a/telicity, and so on, as identified by Vendler 1957; Kenny 1963; Dowty 1979; Krifka 1989, a.o.) are straightforwardly inherited from properties of these models.

Ultimately, we argue that a proper analysis of telicity requires a conception of partitive event structure which is based on language-independent knowledge of the typical (causal) relationships between a range of properties, point events, and states. The success of this approach in explicating imperfective paradox data is thus presented as a motivating example. While this paper, in focusing on telic progressives, necessarily centers the role of causal models in the lexical representation of goal-oriented and durative eventualities, the approach outlined here forms part of a broader project in which causal models form the core of event conceptualization and event-structural relationships. Looking ahead, then, this paper lays the groundwork for a theory on which predicates of eventualities labels for collections of causal conditions which enable, allow, constitute, or otherwise lead to particular outcomes or states, and the familiar aspectual class properties (cf. Vendler 1957; Krifka 1989, among others) are inherited from properties of these causal models.

Appendix: Conceptualizing *process*

As noted in Section 2, one way of thinking about the tension between different approaches to the imperfective paradox is as offering different answers to the following question: how should the *process* component of a complex (telic) event be characterized? This question underlies much of the preceding discussion, but the paper has operated against the background of multiple possible

[37]For instance, while atelic predicates might not have lexically-specified culmination conditions, they might well specify conditions of *termination*, making them compatible with our proposal (32)/(33) for the progressive operator.

conceptions of what a process is, without stating a clear position. This appendix addresses the concept of *process* directly: we make explicit the differences between competing approaches to the nature of processes, and conclude by articulating the specific view that underlies our own analysis.

The literature offers two principled ways of conceptualizing *process*:

(A) **Ontological approach:** Following Stout (1997), this framework treats processes and events as distinct ontological kinds. A process, picked out by a nominalization such as *Mahler's composing the symphony*, does not include the completed outcome, whereas a *completed event* or culmination (picked out by the alternative nominalization *Mahler's composing of the symphony*) constitutes a fully bounded event. The progressive aspect directly picks out the process entity, while the perfective refers to the culmination-bearing event.

(B) **Mereological (part-whole) relationship between events:** Following Landman (1992) and Bonomi (1997), a process can be analyzed in terms of a mereological (part-whole) relationship between events, whereby the process consists of sub-events that collectively instantiate the full eventuality denoted by the predicate. The key issue is determining which sub-events genuinely count as parts of the larger culminating event.

Our view aligns broadly with the mereological perspective, but since we replace the traditional approach to telic predicates' denotations with that of a type-level *causal model* (see Section 3), we do not need to talk about mereological relationships between events. In our framework, a *process* is neither a separate ontological type nor merely a partial stage: it is a *causal assembly of states and events* that aligns with one of the causally sufficient pathways to the culmination C_P. Each state or event in the process is *necessary* within at least one causal pathway for C_P, and collectively they are *sufficient* for reaching C_P under normal conditions (see Section 3.1 for formal definitions). As a result, *processes* can be instantiated (or fail to be instantiated) in particular situations: specifically, a reference-time situation verifies the progressive with respect to P (i.e., verifies that P is *in progress*) just in case it instantiates part of a process for C_P, satisfying some—but not all—conditions from a causally sufficient set, while failing to satisfy a sufficient set for non-culmination.

From the causal modeling perspective, a process for culmination condition C_P is defined as a causally structured set of conditions that are necessary (within some pathways) and jointly sufficient (under normal conditions) for C_P. This formalization allows us to define processes causally and compositionally (see Section 4), without relying on projected culmination or intensional semantics. By replacing the notion of a *mereological relationship between events* with the structure of a causal model, we integrate partitivity and causal adequacy into our definition of process.

References

Abusch, Dorit. 1985. *On Verbs and Time*. Ph. D.thesis, University of Massachusetts, Amherst.

Alonso-Ovalle, Luis and Hsieh, Henrison. 2017. Causes and expectations: on the interpretation of the Tagalog ability/involuntary action form. In Dan Burgdorf, Jacob Collard, Sireemas Maspong and Brynhildur Stefánsdóttir (eds.), *Proceedings of Semantics and Linguistic Theory 27*, pages 75–94.

Altshuler, Daniel. 2014. A typology of partitive aspectual operators. *Natural Language and Linguistic Theory* 32, 735–775.

Arunachalam, Sudha and Kothari, Anubha. 2011. An experimental study of Hindi and English perfective interpretation. *Journal of South Asian Linguistics* 4, 463–508.

Asher, Nicholas. 1992. A default, truth conditional semantics for the progresive. *Linguistics and Philosophy* 15, 463–508.

Bach, Emmon. 1986. The algebra of events. *Linguistics and Philosophy* 9, 5–16.

Baglini, Rebekah and Bar-Asher Siegal, Elitzur. 2025. Modelling Linguistic Causation. *Linguistics and Philosophy* page forthcoming.

Baglini, Rebekah and Francez, Itamar. 2016. The implications of managing. *Journal of Semantics* 33(3), 541–560.

Bar-Asher Siegal, Elitzur. 2025. Causation and Causal Relations: Insights from Natural Language. *Annual Review of Linguistics* 12, forthcoming.

Bar-Asher Siegal, Elitzur, Bassel, Noa and Hagmayer, York. 2021. Causal selection: the linguistic take. In *Experiments in Linguistic Meaning*, volume 1, pages 27–38.

Bar-Asher Siegal, Elitzur and Boneh, Nora. 2020. Causation: from metaphysics to semantics and back. In Elitzur Bar-Asher Siegal and Nora Boneh (eds.), *Perspectives on Causation: selected papers from the Jerusalem 2017 workshop*, pages 3–51, Cham, Switzerland: Springer.

Baratella, Riccardo. 2023. Processes and their modal profile. *Synthese* 201(74).

Beavers, John and Koontz-Garboden, Andrew. 2012. Manner and result in roots of verbal meaning. *Linguistic Inquiry* 43, 331–369.

Belnap, Nuel and Perloff, Michael. 1988. Seeing to it that: a canonical form for agentives. *Theoria* 54, 175–199.

Bjorndahl, Adam and Snider, Todd. 2015. Informative counterfactuals. In Sarah D'Antonio, Mary Moroney and Carol Rose Little (eds.), *Proceedings of Semantics and Linguistic Theory 25*, pages 1–17.

Bonomi, Andrea. 1997. The progressive and the structure of events. *Journal of Semantics* 14, 173–205.

Ciardelli, Ivano, Zhang, Linmin and Champollion, Lucas. 2018. Two switches in the theory of counterfactuals: a study of truth conditionality and minimal change. *Linguistics and Philosophy* 41, 577–621.

Copley, Bridget. 2009. *The semantics of the future*. New York: Routledge.

Copley, Bridget. 2021. Reconciling modal and causal representations of Salish out of control forms. In *Proceedings of the West Coast Conference on Formal Linguistics 39*, page to appear.

Copley, Bridget and Wolff, Philip. 2014. Theories of causation should inform linguistic theories and vice versa. In Bridget Copley and Fabienne Martin (eds.), *Causation in Grammatical Structures*, pages 11–57, Oxford: Oxford University Press.

Davidson, Donald. 1967. The logical form of action sentences. In Nicholas Rescher (ed.), *The Logic of Decision and Action*, pages 81–120, Pittsburgh: University of Pittsburgh Press.

Dowty, David. 1977. Towards a semantic analysis of verb aspect and the English imperfective progressive. *Linguistics and Philosophy* 1, 45–77.

Dowty, David. 1979. *Word Meaning and Montague Grammar*. Dordrecht: Reidel.

Dowty, David. 1991. Thematic Proto-Roles and Argument Selection. *Language* 67, 547–619.

Grimshaw, Jane. 2005. Semantic structure and semantic content in lexical representation. In Jane Grimshaw (ed.), *Words and Structure*, Stanford, CA: CSLI Publications.

Hale, Ken and Keyser, Samuel Jay. 2002. *Prolegomenon to a Theory of Argument Structure*. Cambridge, MA: MIT Press.

Halpern, Joseph Y. 2015. A modification of the Halpern-Pearl Definition of Causality. In Qiang Yang and Michael Wooldridge (eds.), *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 24)*, pages 3022–3033, Palo Alto, CA: AAAI Press.

Halpern, Joseph Y. and Pearl, Judea. 2005. Causes and explanations: A structural-model approach – Part I: Causes. *British Journal of Philosophy of Science* 56, 843–887.

Hausman, Daniel M. 1998. *Causal Asymmetries*. New York: Cambridge University Press.

Hausman, Daniel Murray. 2005. Causal Relata: Tokens, Types, or Variables? *Erkenntnis* 63(1), 33–54.

Henderson, Robert. 2010. Non-defeasible counterfactuality blocks epistemic inference: evidence from "if not for" counterfactuals, ms, University of California, Santa Cruz.

Hitchcock, Christopher. 2020. Communicating causal structure. In Elitzur Bar-Asher Siegal and Nora Boneh (eds.), *Perspectives on causation: selected papers from the Jerusalem 2017 workshop*, Jerusalem Studies in Philosophy and History of Science, pages 53–71, Springer Cham.

Hobbs, Jerry. 2005. Toward a useful concept of causality for lexical semantics. *Journal of Semantics* 22(2), 181–209.

Hoover, Kevin. 2001. *Causality in Macroeconomics*. Cambridge: Cambridge University Press.

Hume, David. 1748. *An Enquiry Concerning Human Understanding*. Oxford: Oxford University Press.

Kaufmann, Stefan. 2013. Causal premise semantics. *Cognitive Science* 37, 1136–1170.

Kenny, Anthony. 1963. *Action, Emotion, and Will: Studies in Philosophical Psychology*. London: Routledge and Kegan Paul.

Klein, Wolfgang. 1994. *Time in Language*. New York: Routledge.

Koenig, Jean-Pierre and Muansuwan, Nuttanart. 2000. How to end without ever finishing: Thai semi-perfectivity. *Journal of Semantics* 17, 147–182.

Kratzer, Angelika. 1981. The Notional Category of Modality. In Hans-Jürgen Eikmeyer and Hannes Rieser (eds.), *Words, Worlds, and Contexts. New Approaches in Word Semantics*, pages 38–74, Berlin: de Gruyter.

Kratzer, Angelika. 1998. More structural analogies between pronouns and tense. In Devon Strolovich and Aaron Lawson (eds.), *Proceedings of Semantics and Linguistic Theory 8*, pages 92–110, Ithaca, NY: CLC Publications, Cornell University.

Kratzer, Angelika. 2004. Telicity and the meaning of objective case. In Jacqueline Guéron and Jacqueline Lecarme (eds.), *The Syntax of Time*, pages 389–423, Cambridge, MA: MIT Press.

Krifka, Manfred. 1989. Nominal reference and quantification in event semantics. In Renate Bartsch, Johan van Benthem and Peter van Emde Boas (eds.), *Semantics and Contextual Expression*, pages 75–115, Dordrecht: Foris Publications.

Landman, Fred. 1992. The progressive. *Natural Language Semantics* 1, 1–32.

Lascarides, Alex. 1991. The progressive and the imperfective paradox. *Synthese* 87, 401–447.

Levin, Beth and Rappaport Hovav, Malka. 1991. Wiping the slate clean: a lexial semantic exploration. *Cognition* 41, 123–151.

Levin, Beth and Rappaport Hovav, Malka. 1999. Two Structures for Compositionally Derived Events. In Tanya Matthews and Devon Strolovitch (eds.), *Semantics and Linguistic Theory*, volume 9, pages 199–223.

Levin, Beth and Rappaport Hovav, Malka. 2005. *Argument realization*. Cambridge: Cambridge University Press.

Levin, Beth and Rappaport Hovav, Malka. 2013. Lexicalized meaning and manner/result complementarity. In Boban Arsenijević, Berit Gehrke and Rafael Marín (eds.), *Subatomic Semantics of Event Predicates*, pages 49–70, Dordrecht: Springer.

Levin, Beth and Rappaport Hovav, Malka. 2014. Manner and result: a view from *clean*. In Rob Pensalfini, Myfany Turpin and Damian Guillemi (eds.), *Language Description Informed by Theory*, pages 337–357, Amsterdam: John Benjamins.

Lewis, David. 1973a. Causation. *Journal of Philosophy* 70(17), 556–567.

Lewis, David. 1973b. *Counterfactuals*. Oxford: Blackwell.

Lewis, David. 1979. Counterfactual dependence and time's arrow. *Noûs* 13, 455–476.

Mackie, John. 1965. Causes and conditions. *American Philosophical Quarterly* 2(4), 245–265.

Martin, Fabienne. 2018. Time in probabilistic causation: direct vs. indirect uses of lexical causative verbs. In *Proceedings of Sinn und Bedeutung 22*.

Martin, Fabienne. 2019. Non-culminating accomplishments. *Language and Linguistics Compass* 13, 1–20.

Moens, Mark and Steedman, Mark. 1988. Temporal ontology and temporal reference. *Computational Linguistics* 14, 15–28.

Mourelatos, Alexander P. D. 1978. Events, processes, and states. *Linguistics and Philosophy* 2(3), 415–434.

Nadathur, Prerna. 2023a. *Actuality inferences: causality, aspect, and modality*, volume 15 of *Oxford Studies in Semantics and Pragmatics*. Oxford: Oxford University Press.

Nadathur, Prerna. 2023b. Causal semantics for implicative verbs. *Journal of Semantics* to appear.

Nadathur, Prerna and Filip, Hana. 2021. Telicity, teleological modality, and (non-)culmination. In *Proceedings of the West Coast Conference on Formal Linguistics*, volume 39, page to appear.

Nadathur, Prerna and Lauer, Sven. 2020. Causal necessity, causal sufficiency, and the implications of causative verbs. *Glossa,* 5, 49.1–37.

Parsons, Terence. 1989. The progressive in English: events, states, and processes. *Linguistic and Philosophy* 12, 213–241.

Parsons, Terence. 1990. *Events in the Semantics of English: A Study in Subatomic Semantics*, volume 19 of *Current Studies in Linguistics*. Cambridge, MA: MIT Press.

Partee, Barbara. 1973. Some structural analogies between tenses and pronouns. *The Journal of Philosophy* pages 601–609.

Pearl, Judea. 2000. *Causality*. Cambridge: Cambridge University Press.

Rappaport Hovav, Malka. 2017. Grammatically Relevant Ontological Categories Underlie Manner/Result Complementarity. In *Proceedings of Israel Association for Theoretical Linguistics 32*.

Reichenbach, Hans. 1947. *Elements of Symbolic Logic*. New York: The Free Press.

Rothstein, Susan. 2000. Secondary predication and aspectual structure. In *Approaching the Grammar of Adjuncts*, volume 17 of *ZAS Papers in Linguistics*, pages 241–264.

Schulz, Katrin. 2011. If you'd wiggled A, then B would've changed. *Synthese* 179, 239–251.

Singh, Mona. 1998. On the semantics of the perfective aspect. *Natural Language Semantics* 6(171–199).

Sosa, Ernest and Tooley, Michael. 1993. *Causation*. Oxford: Oxford University Press.

Spirtes, Peter, Glymour, Clark and Scheines, Richard. 1993. *Causation, Prediction, and Search.* New York: Springer Verlag.

Steward, Helen. 2013. Processes, Continuants, and Individuals. *Mind* 122(487), 781–812.

Steyvers, Mark, Tenenbaum, Joshua, Wagenmakers, Eric-Jan and Blum, Ben. 2003. Inferring causal networks from observations and interventions. *Cognitive Science* 27, 453–489.

Stout, Rowland. 1997. Processes. *Philosophy* 72(279), 19–27.

Szabó, Zoltan. 2004. On the progressive and the perfective. *Noûs* 38, 29–59.

Szabó, Zoltan. 2008. Things in progress. *Philosophical Perspectives* 22, 499–525.

Thomason, Richmond. 1984. Combinations of tense and modality. In Dov Gabbay and Franz Guenther (eds.), *Handbook of Philosophical Logic: Extension of Classical Logic*, pages 135–165, Dordrecht: Reidel.

Tooley, Michael. 1987. *Causation: A Realist Approach*. Oxford: Oxford University Press.

Varasdi, Karoly. 2014. Making progressives: necessary conditions are sufficient. *Journal of Semantics* 31, 179–207.

Vendler, Zeno. 1957. Verbs and times. *The Philosophical Review* 66, 143–160.

Vlach, Frank. 1981. The semantics of the progressive. In Philip Tedeschi and Annie Zaenen (eds.), *Syntax and semantics 14: Tense and aspect*, pages 271–294, New York: Academic Press.

von Fintel, Kai and Iatridou, Sabine. 2005. What to do if you want to go to Harlem, ms, Massachusetts Institute of Technology.

Woodward, James. 2003. *Making Things Happen: A Theory of Causal Explanation*. Oxford: Oxford University Press.

Zucchi, Sandro. 1999. Incomplete events, intensionality, and imperfective aspect. *Natural Language Semantics* 7, 179–215.